Where Does the Gain in Dynamic Convolution Come From?

A controlled study of routing, capacity and width on two backbones*

Ankit Kumar

Mechatronics Department, University of Agder, Grimstad, Norway

24 September 2026; revised 17 October 2026

Abstract

Dynamic convolution replaces one kernel with a bank of K kernels mixed by input-dependent weights. Its gains are usually credited to input-adaptive kernel selection, but published comparisons set it against a network with K times fewer parameters, so routing and capacity are confounded. We separate them with a constant mixture of the same bank, a router fed another image's descriptor and a router frozen at random initialisation, on ResNet-18 and MobileNetV2, CIFAR-100 and Tiny-ImageNet, across 264 training runs with six seeds each and equivalence tests at a ± 0.5 pp margin. Seven findings. (1) The kernel bank on its own adds nothing measurable: a uniformly mixed bank is never significantly different from a plain network and is equivalent to one in 7 of ten backbone and condition pairs; the other three, all on MobileNetV2, are inconclusive. (2) Wherever dynamic convolution beats the plain network, which it does in 9 of the ten pairs, input-dependent mixing carries the effect; the exception, ResNet-18 on CIFAR-100 at width 64, is a well-powered null that we could not explain. (3) On MobileNetV2 the mixing does not have to be learned: a router frozen at random initialisation is not distinguishable from a trained one, while routing each image with another image's descriptor removes the benefit. (4) Banks larger than $K = 2$ are not distinguishable from $K = 2$. (5) Whether routing is worth its cost depends on the cost held fixed and on the backbone: at equal parameters a plain ResNet-18 of twice the width beats the routed one by 2.3 to 4.3 pp, while on MobileNetV2 a plain network with 1.7 to 2.0 times the parameters wins only one of three pairs; at equal width, for at most 8 % more compute, routing beats the plain network in 4 of five conditions on ResNet-18 and 5 of five on MobileNetV2. (6) Annealing the softmax temperature, as DY-Conv recommends, never improved accuracy and lowered it in 4 of six conditions. (7) Routing entropy is not a reliable measure of whether routing helps: on CIFAR-100 at width 64 the trained routers of the two backbones have nearly the same entropy and improve on the constant mixture by +0.03 and +1.47 pp. Of four predictions recorded before the experiments that test them, three failed; we report each one.

1 Introduction

A dynamic convolution layer stores K kernels and combines them per input. For an input feature map F and kernel bank $\{\mathcal{W}_k\}_{k=1}^{K}$, the layer computes

$$\mathcal{W}_{\text{dyn}}(F) = \sum_{k=1}^{K} z_k(F)\,\mathcal{W}_k, \qquad \mathbf{z} = \operatorname{softmax}\!\left(\frac{g(\phi(F))}{\tau}\right), \tag{1}$$

where ϕ summarises the feature map, g is a small multilayer perceptron and τ is a softmax temperature. Nearly every published variant sets ϕ to global average pooling (Yang et al., 2019; Chen et al., 2020).

The usual account of why this helps is that $\mathbf{z}$ adapts the kernel to the input. A second account is available and is rarely tested. The bank multiplies the stored convolution parameters by K, and published comparisons are made against a network with K times fewer parameters. Under that comparison, a gain from routing and a gain from extra capacity look identical.

This report separates the two. We compare the standard routed layer against a constant mixture that has the same bank and no routing, a router fed a different image's descriptor, a router frozen at random initialisation, and a plain network. The comparison spans three widths, three bank sizes, two temperature schedules, two datasets and two backbones: ResNet-18, where almost every parameter sits in a dynamic layer, and MobileNetV2, where the dynamic layers are depthwise and hold 3 % of the parameters. That is 264 training runs and 458 GPU-hours on one RTX 4070 Ti. Every comparison is paired by seed and tested for equivalence as well as for difference, so that a null result is a claim with a stated margin rather than a failure to reject. The two backbones run the same program, and the MobileNetV2 predictions were recorded, dated, before its first run.

Terms. Throughout, *routing* means that the mixing weights $\mathbf{z}$ depend on the input image, whether or not the function that produces them is trained. A *trained router* is one whose weights are optimised. The *bank term* is the accuracy difference `uniform` − `static`, the effect of the extra kernels with no routing; the *routing term* is `gap` − `uniform`, the effect of routing the same bank.

*Draft of 17 October 2026. ResNet-18 and MobileNetV2 are complete; the same program on VGG-16, ResNet-50 and ConvNeXt-Tiny is pre-registered and running.

What we find. Each result section reports both backbones side by side, and Section 13 collects the results in one table.

1. **The kernel bank on its own adds nothing measurable** (both backbones; Section 5). The bank term is not significant in any of the ten backbone and condition pairs, and is equivalent to zero in 7 of them; the other three, all on MobileNetV2, are inconclusive.
2. **Wherever dynamic convolution helps, routing carries the effect; where it helps depends on the backbone** (Sections 5 and 6). Dynamic convolution beats the plain network in 9 of ten pairs, and the routing term is significant in 9: 4 of five on ResNet-18 and 5 of five on MobileNetV2. The exception is ResNet-18 on CIFAR-100 at width 64, where no arm differs from the plain network. That null is well powered, two explanations we tested failed, and MobileNetV2 shows +1.360 pp in the same condition.
3. **On MobileNetV2 routing does not have to be learned** (MobileNetV2 only; Section 7). A router frozen at random initialisation is not distinguishable from a trained one, and routing each image with another image's descriptor removes the benefit. On ResNet-18 the test is uninformative, because in the one condition where it was run no arm differs from the plain network.
4. **Banks larger than $K = 2$ are not distinguishable from $K = 2$** (both; Section 8). On ResNet-18 they are equivalent to it; on MobileNetV2 the comparison is inconclusive.
5. **Whether routing is worth its cost depends on the cost held fixed and on the backbone** (Section 9). At equal parameters a plain ResNet-18 of twice the width beats the routed one in all three pairs; a plain MobileNetV2 with 1.7 to 2.0 times the routed network's parameters wins one of three. At equal width, where routing adds at most 8 % to the multiply-accumulates, routing wins 4 of five conditions on ResNet-18 and 5 of five on MobileNetV2.
6. **Annealing the softmax temperature never improved accuracy** (both; Section 10). It lowered accuracy in 4 of six conditions, at all three MobileNetV2 widths by about 1 pp and on ResNet-18 at width 32, and made no detectable difference in the other two.
7. **Routing entropy is not a reliable measure of whether routing helps** (Section 11). Within ResNet-18 it does not order the benefit; within MobileNetV2 it does; across the two backbones the same entropy accompanies benefits of +0.03 and +1.47 pp. After annealing, routers end close to uniform: on MobileNetV2 in 15 to 16 of its 17 dynamic layers.

Predictions. We recorded four predictions before running the experiments that test them, and three failed (Section 12). On ResNet-18 we attributed the CIFAR-100 width-64 null to the network memorising its training set and predicted that the effect would vanish on Tiny-ImageNet at width 64; it did not. For MobileNetV2 we predicted that the bank term would be equivalent to zero in all five conditions, that width would beat routing in all three equal-width pairs, and that routing would help in at least three conditions. Only the last held. The first failed because three bank terms are inconclusive, not because any is significant, so it does not contradict finding 1.

2 Related work

Dynamic convolution. CondConv (Yang et al., 2019) and DY-Conv (Chen et al., 2020) introduced input-dependent mixtures of kernels. ODConv (Li et al., 2022) extends the attention across several kernel dimensions, and KernelWarehouse (Li and Yao, 2024) reorganises the bank so that far larger K becomes affordable. The reported gains are large: for a ResNet-18 on ImageNet at four times the kernel parameters, Li and Yao (2024) report top-1 improvements of 3.38 pp for DY-Conv, 4.01 pp for ODConv and 5.61 pp for KernelWarehouse under their training recipe. FDConv (Chen et al., 2025) observes that the parallel kernels of DY-Conv tend toward highly similar frequency responses, so that the extra parameters buy little adaptability. SD-Conv (He et al., 2023) prunes half the parameters of a trained dynamic network at a cost below 1 pp on CIFAR-100, and Li et al. (2021) argue that the K static kernels are redundant. To our knowledge none of these works isolates the bank from the routing with a constant mixture of the same bank. Our finding that the bank contributes nothing on its own is consistent with these observations. Spatially resolved variants such as DRConv (Chen et al., 2021) route on local rather than global signals and fall outside the controls used here.

Optimisation difficulty. Li et al. (2021) identify the joint optimisation of dynamic attention and static kernels as the central difficulty of dynamic convolution and propose a matrix decomposition that eases it. DY-Conv addresses the same difficulty by annealing the softmax temperature from 30 to 1 early in training (Chen et al., 2020). Section 10 tests that schedule.

Re-parameterisation. A constant mixture of K kernels is a linear re-parameterisation of a single kernel. Such re-parameterisations can change optimisation without changing the function class (Arora et al., 2018), and ACNet and RepVGG exploit this by training a richer block and merging it into one kernel at inference (Ding et al., 2019, 2021). Our `uniform` arm is exactly such a mergeable block.

Memorisation. Networks of the size used here can fit their training sets completely, even under random labels (Zhang et al., 2017). That motivated the saturation hypothesis tested in Section 6.

Equivalence testing. The two one-sided tests procedure (Schuirmann, 1987) asserts that a difference lies within a stated margin; Lakens (2017) give a practical treatment. It is standard in bioequivalence studies and rare in this literature, where a large p-value from a difference test is often read as evidence of no effect.

3 Method

3.1 Routing signals

All arms share Eq. (1), the backbone, the optimiser and the seeds, and differ only in how $\mathbf{z}$ is produced. Write $\mu(F) \in \mathbb{R}^C$ for the channel-wise global average.

`static` No dynamic layer: the plain backbone, $K = 1$.

`uniform` $g \equiv \mathbf{0}$, so $\mathbf{z} = (1/K, \ldots, 1/K)$ for every input. The bank exists and is trained; nothing is routed.

`gap` $\mathbf{z} = \mathrm{softmax}(g(\mu(F))/\tau)$ with g trained. The standard layer.

`shuffled` As `gap`, but sample i is routed with the descriptor of sample $\pi(i)$ for a permutation π redrawn on every forward pass, at evaluation as well as training. The distribution of $\mathbf{z}$ is preserved and the pairing between image and kernel is destroyed.

`frozen_random` As `gap`, with every weight of g frozen at its random initialisation. Routing varies with the input but is never optimised.

The five main conditions (CIFAR-100 at widths 16, 32 and 64, Tiny-ImageNet at widths 32 and 64) each have `static`, `uniform` and `gap`. `shuffled` and `frozen_random` were run on CIFAR-100 at width 64 only. The bank-size groups ($K = 2$ and $K = 8$) and the temperature groups run `gap` only, at CIFAR-100 widths.

3.2 The constant mixture is a single kernel

Proposition 1. *If $\mathbf{z} = (1/K, \ldots, 1/K)$ for every input, the layer computes $F * \bar{\mathcal{W}}$ with $\bar{\mathcal{W}} = \frac{1}{K}\sum_k \mathcal{W}_k$, and its effective parameter count equals that of one static kernel.*

Proof. The mixture in Eq. (1) is constant, so $\mathcal{W}_{\text{dyn}} = \bar{\mathcal{W}}$ for every F. □

At inference `uniform` is therefore the plain network. On ResNet-18 at width 64 it stores 44.18M parameters and has an effective count of 11.22M. It spans the same function class as `static`, so any difference between the two is an effect of optimisation. That makes it the right control for the extra parameters of the bank.

3.3 Two exact implementations

Convolution is linear in its weights, so

$$F * \Big(\sum_k z_k \mathcal{W}_k \Big) = \sum_k z_k \, (F * \mathcal{W}_k). \tag{2}$$

The left side builds a per-sample kernel and runs one grouped convolution; on ResNet-18 at width 64 and batch 128 its aggregated weights alone take about 1.2 GB per stage-4 layer. The right side runs K ordinary convolutions and mixes the outputs, which costs K times the arithmetic but no per-sample weights. Every run uses the right side. The test suite asserts that the two agree to 10^{-10} in double precision across four routing signals.

3.4 Training protocol

Data. CIFAR-100 (Krizhevsky, 2009), 32×32, 100 classes, at widths 16, 32 and 64. Tiny-ImageNet (Le and Yang, 2015), 64×64, 200 classes, at widths 32 and 64. The stem is the same on both: a 3×3 stride-1 convolution with no max-pool. At 64×64 this leaves an 8×8 final feature map rather than 4×4 and roughly quadruples the cost of the deeper stages, but changing the stem would change the network between datasets. Crop padding is 4 pixels at 32 and 8 at 64.

Recipe. SGD with Nesterov momentum 0.9, learning rate 0.1, weight decay 5×10^{-4}, batch 128, five warm-up epochs then cosine decay over 150 epochs. Augmentation is random cropping and horizontal flipping only. Mixed precision in bfloat16. Six seeds, 0 to 5, shared by every arm. $K = 4$ unless stated. Both backbones use this recipe unchanged, and every group of both trained at batch 128 (on ResNet-18 at Tiny-ImageNet width 64 after a one-epoch memory probe, so the two Tiny-ImageNet widths are directly comparable).

ResNet-18. ResNet-18 (He et al., 2016) in which all 16 3×3 convolutions inside residual blocks are dynamic; the stem and the 1×1 shortcuts stay static.

MobileNetV2. MobileNetV2 (Sandler et al., 2018), adapted to small inputs in the usual way: the stem and the second stage use stride 1. Width 64 is the standard network; widths 32 and 16 scale every channel count by $1/2$ and $1/4$, rounded to a multiple of 8. The rule that decides which layers are dynamic is the ResNet-18 rule without change: every $k \times k$ convolution with $k > 1$ inside a block. In MobileNetV2 that is the 17 depthwise 3×3 convolutions; the stem, every 1×1 expansion and projection, and the classifier stay static. The consequences differ from ResNet-18. At CIFAR-100 width 64 the bank adds 0.19M parameters to a 2.35M network, so only 3 % of the static parameters sit in dynamic layers, and the router of the `gap` arm (the ResNet-18 design, a two-layer perceptron of hidden width $C_{\text{in}}/4$) has 1137K parameters, 48 % of the static network and more than the bank it routes. Static convolutions and every bank kernel share one initialisation, Kaiming-normal with fan-out.

Temperature. $\tau = 1$ throughout, except in the temperature ablation (Section 10), where τ falls linearly from 30 to 1 over the first ten epochs and stays at 1 for the remaining 140.

3.5 Routing entropy

The normalised routing entropy $\tilde{H} = H(\mathbf{z})/\log K$ is 1 for a uniform mixture and 0 for a one-hot one; we report $1 - \tilde{H}$, the departure from uniform. Each run records it once, after training, on the last batch of the unshuffled test set, which is the same 16 images in every run on a dataset. A reported value is the mean over those images and over the dynamic layers. It was not recorded during training, and the trained weights were not kept, so it cannot be recomputed on the full test set (Section 14). No accuracy depends on it.

3.6 Coverage and provenance

All 132 ResNet-18 runs and all 132 MobileNetV2 runs finished; none diverged. The first MobileNetV2 run, a recorded gate, reached 72.66 % on CIFAR-100 against a floor of 65 %. One MobileNetV2 run (Tiny-ImageNet, width 64, `gap`, seed 1, final accuracy 61.15 %) was stopped by a GPU fault at epoch 125 and resumed from its checkpoint; without it, `gap` − `static` in that condition is +1.526 pp ($p = 0.0007$, $n = 5$), so no conclusion depends on it. MobileNetV2 accuracies and training losses are parsed from the training log and checked against the 132 result files: every final accuracy is identical, and every final training loss agrees to the three decimals the log prints. Routing entropies are read from the result files. The appendix lists every run (Figures 17 and 18, Tables 19, 24, 23 and 28) and every comparison against `static` (Tables 20 and 25).

4 Statistical protocol

Equivalence. Several conclusions are null conclusions, and a large p from a difference test does not establish one. For paired differences d_i with mean $\bar{d}$ and standard error $s_{\bar{d}}$ we use two one-sided tests against a margin δ,

$$p_{\text{TOST}} = \max\left\{1 - T_{n-1}\Big(\tfrac{\bar{d}+\delta}{s_{\bar{d}}}\Big),\ T_{n-1}\Big(\tfrac{\bar{d}-\delta}{s_{\bar{d}}}\Big)\right\}, \tag{3}$$

and read $p_{\text{TOST}} < 0.05$ as a positive claim that the true difference lies within $\pm\delta$. We fixed $\delta = 0.5\,\text{pp}$ before the first run. The pre-registration justified this by reference to reported gains at CIFAR scale without citing a source. The nearest documented figures we have since found are the ImageNet ResNet-18 gains of 3.38 to 5.61 pp quoted above, and the margin sits an order of magnitude below them.

Why six seeds. The two-sided Wilcoxon signed-rank test (Wilcoxon, 1945) on n pairs admits only 2^n equally likely sign assignments, so its smallest attainable p is 2^{1-n}: 0.25 at $n = 3$, 0.0625 at $n = 5$, 0.031 at $n = 6$. Six is the smallest count at which it can reach 0.05. We report it beside the paired t-test throughout.

Power. For `gap − uniform` on CIFAR-100 at width 64 the paired standard deviation is 0.257 pp, giving a minimum detectable difference of 0.270 pp at $n = 6$. Simulated power (2×10^5 draws) is 48.5 % for a true effect of 0.25 pp, 96.3 % at 0.5 pp and 100.0 % at 1.0 pp. The ResNet-18 null on CIFAR-100 at width 64 (Section 6) is not a power artefact.

Pre-registration. Arms, recipe, primary metric, seeds, margin and the decision table were written down before the first run finished. The primary metric is final-epoch top-1 accuracy, since best-epoch accuracy maximises over a noisy sequence. The Tiny-ImageNet width-64 prediction of Section 6 was recorded in writing, dated, before any of those runs. An expectation about Tiny-ImageNet width 32 was stated before those runs but not recorded; we give it correspondingly less weight. For MobileNetV2, three predictions and a gate (the first static run had to reach 65 % on CIFAR-100 before the other 131 runs were spent) were recorded on 30 September 2026, before any MobileNetV2 run (Section 12).

5 The bank adds nothing; routing carries the effect

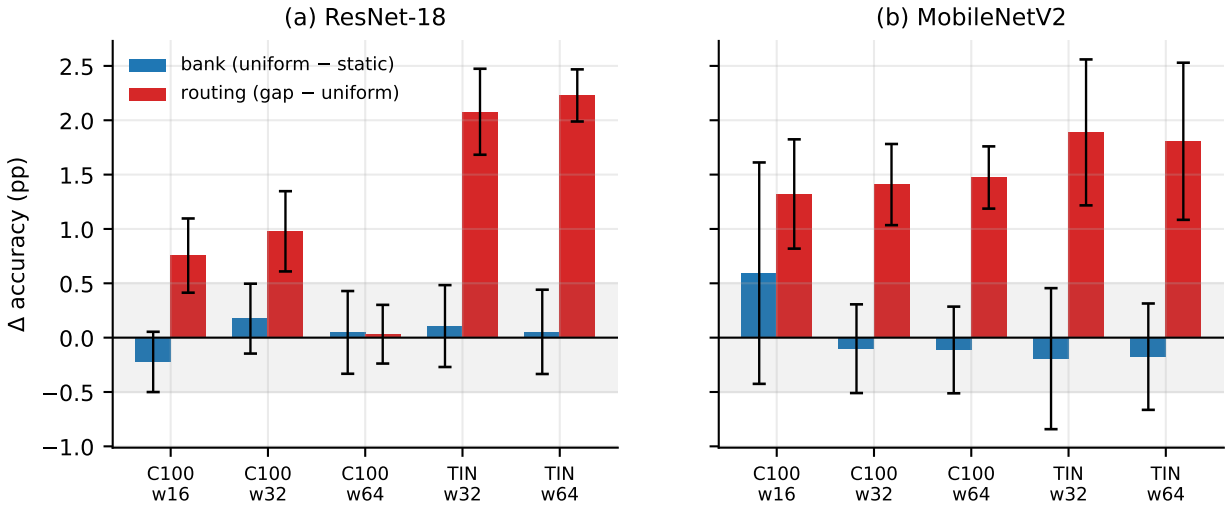


Figure 1: Bank term (`uniform − static`) and routing term (`gap − uniform`) in the five conditions, for (a) ResNet-18 and (b) MobileNetV2, 95 % CIs. The shaded band is the ±0.5 pp margin.

Figure 1 splits the total effect `gap − static` into the bank term and the routing term in the five main conditions of each backbone.

The bank term. On ResNet-18 the bank term is equivalent to zero in all five conditions, with p_{TOST} between 0.014 and 0.025; the largest is +0.175 pp (Table 1). On MobileNetV2 it is never significantly different from zero (Table 2). It is equivalent to zero on CIFAR-100 at widths 32 and 64 ($p_{\text{TOST}} = 0.0269$ and 0.0275). At CIFAR-100 width 16 and at both Tiny-ImageNet widths it is inconclusive: the point estimates (+0.593, -0.193 and -0.175 pp) lie inside the margin, but the confidence intervals do not. At CIFAR-100 width 16 the MobileNetV2 `static` arm itself is unusually variable across seeds (Table 24). Over both backbones the bank term is significant in none of the ten pairs and equivalent to zero in 7. On MobileNetV2 static and bank kernels share one initialisation, so its bank term is free of the initialisation mismatch that affects the ResNet-18 bank term (Section 14), and it gives the same answer.

The routing term. The routing term is positive with $p < 0.05$ in 9 of the ten pairs: in 4 of five on ResNet-18 and in all five on MobileNetV2, where it ranges from 1.32 to 1.89 pp. Because the bank term is nil, the total effect `gap − static` is essentially the routing term: on ResNet-18 at Tiny-ImageNet width 64, for example, routing is +2.228 pp of a +2.282 pp total. At ResNet-18 CIFAR-100 width 16 the bank term is slightly negative (-0.223 pp, equivalent to zero) and the routing term therefore exceeds the total, which is why we report no percentage share.

Where the effect appears. On MobileNetV2 the total effect is roughly flat across width on both datasets, from 1.31 to 1.91 pp (Figure 2). On ResNet-18 it is +0.532 pp at CIFAR-100 width 16 ($p = 0.0023$), +1.153 pp at width 32 ($p < 10^{-4}$) and +0.080 pp at width 64 ($p = 0.6158$), and +2.185 and +2.282 pp on Tiny-ImageNet at widths 32 and 64. The single condition without an effect, ResNet-18 on CIFAR-100 at width 64, is the subject of Section 6. Figures 15 and 16 in the appendix show the ResNet-18 accuracies and decomposition separately.

Table 1: ResNet-18: bank and routing terms, paired by seed. The bank term is equivalent to zero in all five conditions.

		bank (`uniform−static`)			routing (`gap−uniform`)		total	
dataset	width	Δ (pp)	p_{TOST}	verdict	Δ (pp)	p	Δ (pp)	p
CIFAR-100	16	-0.223	0.0251	equivalent	+0.755	0.0023	+0.532	0.0023
CIFAR-100	32	+0.175	0.0242	equivalent	+0.978	0.0010	+1.153	$< 10^{-4}$
CIFAR-100	64	+0.048	0.0142	equivalent	+0.032	0.7752	+0.080	0.6158
Tiny-ImageNet	32	+0.107	0.0218	equivalent	+2.078	$< 10^{-4}$	+2.185	$< 10^{-4}$
Tiny-ImageNet	64	+0.053	0.0157	equivalent	+2.228	$< 10^{-4}$	+2.282	$< 10^{-4}$

Table 2: MobileNetV2: bank and routing terms, paired by seed.

		bank (`uniform−static`)			routing (`gap−uniform`)		total	
dataset	width	Δ (pp)	p_{TOST}	verdict	Δ (pp)	p	Δ (pp)	p
CIFAR-100	16	+0.593	0.5884	inconclusive	+1.322	0.0011	+1.915	0.0225
CIFAR-100	32	-0.102	0.0269	equivalent	+1.408	0.0002	+1.307	0.0015
CIFAR-100	64	-0.113	0.0275	equivalent	+1.473	$< 10^{-4}$	+1.360	0.0002
Tiny-ImageNet	32	-0.193	0.1393	inconclusive	+1.888	0.0008	+1.695	0.0028
Tiny-ImageNet	64	-0.175	0.0742	inconclusive	+1.807	0.0014	+1.632	0.0002

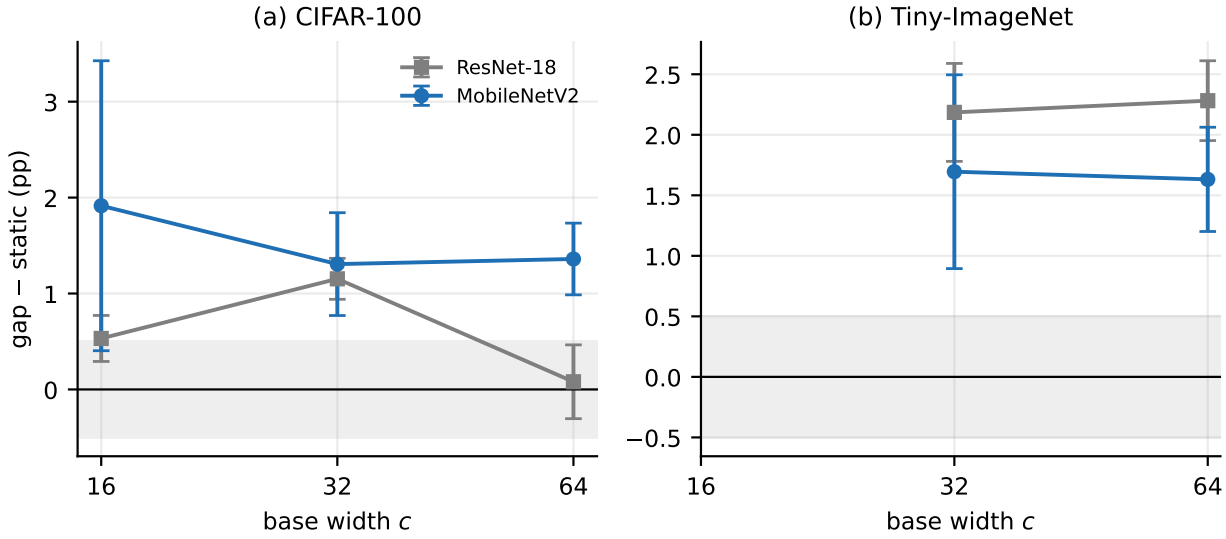


Figure 2: `gap−static` against width for both backbones, 95 % CIs. ResNet-18 has a null on CIFAR-100 at width 64; MobileNetV2 does not.

6 The one null: ResNet-18 on CIFAR-100 at width 64

On ResNet-18 at CIFAR-100 width 64 every arm is equivalent to the plain network (Table 20, Figure 3). The largest gap between arm means is 0.250 pp, against a median within-arm seed standard deviation of 0.238 pp, and a one-way ANOVA across the five arms gives $F(4, 25) = 0.954$, $p = 0.450$. The null is not a power artefact: the minimum detectable routing term is 0.270 pp (Section 4). On MobileNetV2 the same condition behaves like every other: the five arms differ ($F = 42.8$, $p < 10^{-4}$) and `gap − static` is +1.360 pp (95 % CI [+0.986, +1.734], $p = 0.0002$). We tested two explanations of the ResNet-18 null.

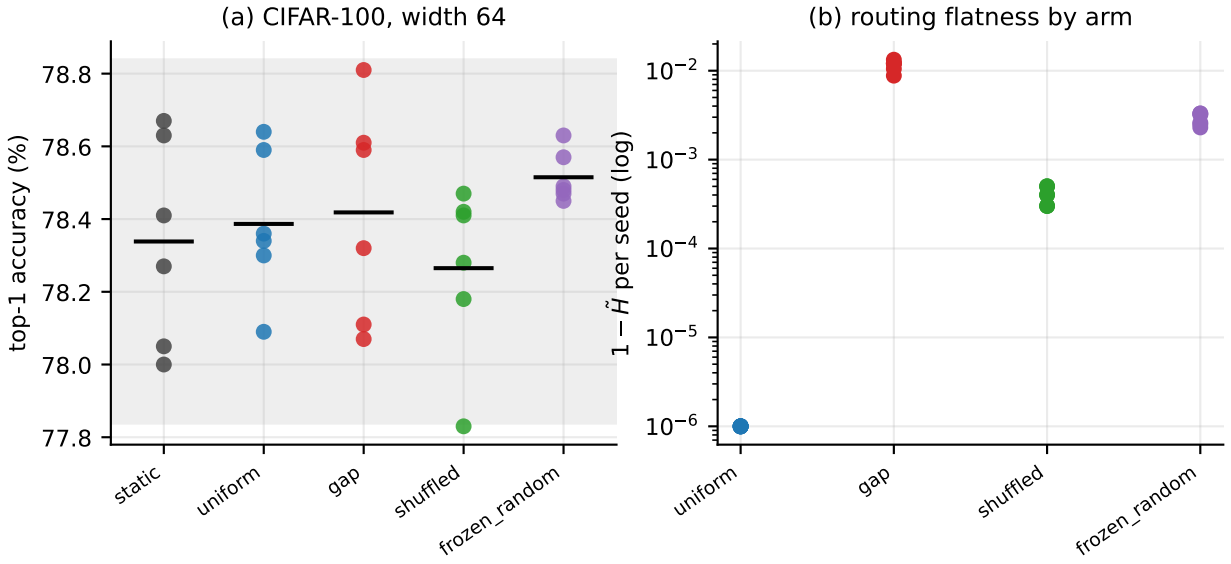


Figure 3: ResNet-18, CIFAR-100, width 64. (a) Every run of the five arms; bars are arm means and the shaded band is ±0.5 pp around `static`. (b) Router flatness $1 - \tilde{H}$ per seed, log scale; `uniform` is exactly zero and is drawn at the axis floor.

6.1 Saturation: a recorded prediction that failed

The saturation account says the routing effect vanishes once a network memorises its training data. On CIFAR-100 the ResNet-18 null coincides with the lowest static training loss, 0.0080.

Tiny-ImageNet, width 32. Here `gap − static` is +2.185 pp (95 % CI [+1.781, +2.589], $p < 10^{-4}$), against +1.153 pp on CIFAR-100 at the same width. Before these runs we expected the effect to be larger than on CIFAR-100, since the static arm trains to a higher loss (0.1194 against 0.0170). That expectation held, but it was stated and not recorded, it predicted only a direction, and we do not treat it as evidence.

Tiny-ImageNet, width 64. Before these runs we recorded, in writing and dated, the prediction in Table 3: the static arm would train to a loss below 0.05, and the effect would then fall below 0.5 pp and be equivalent to zero. The loss condition held (0.0210) and the effect did not vanish: `gap − static` = +2.282 pp, 95 % CI [+1.952, +2.611], $p = 1.0 \times 10^{-5}$ (Figure 4). The prediction failed.

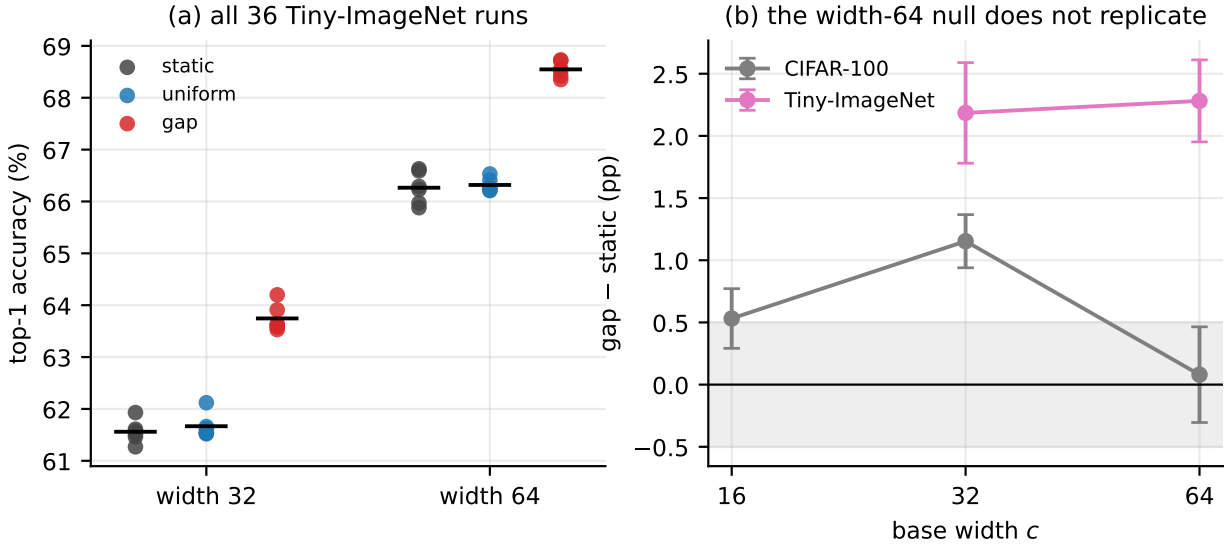


Figure 4: ResNet-18. (a) All 36 Tiny-ImageNet runs. (b) `gap − static` by width on both datasets, 95 % CIs. The CIFAR-100 width-64 null does not appear on Tiny-ImageNet.

Table 3: ResNet-18: the prediction recorded before the Tiny-ImageNet width-64 runs, and the outcome. Both conditions were stated as required.

	prediction	measured	outcome
(a)	`gap` − `static` below 0.5 pp, equivalent by TOST	+2.282 pp, $p = 1.0 \times 10^{-5}$, $p_{\text{TOST}} = 1.0000$	failed
(b)	static final training loss below 0.05	0.021	held

Training loss does not order the effect. Ordered by the static arm's final training loss (Table 4, Figure 5b), the ResNet-18 effects do not follow: Tiny-ImageNet at width 64 trains to a loss close to CIFAR-100 at width 32 and shows twice the effect, and CIFAR-100 at width 16 trains to the highest loss and shows one of the smallest. MobileNetV2 is consistent with the saturation account but cannot rescue it (Figure 6): none of its static arms trains below 0.248 under this recipe, and every MobileNetV2 condition has an effect. The ResNet-18 Tiny-ImageNet width-64 point, at a loss of 0.0210 with one of the largest effects, still rules out a single loss threshold for all ten points, and a threshold chosen after seeing them would not be evidence. We do not adopt it.

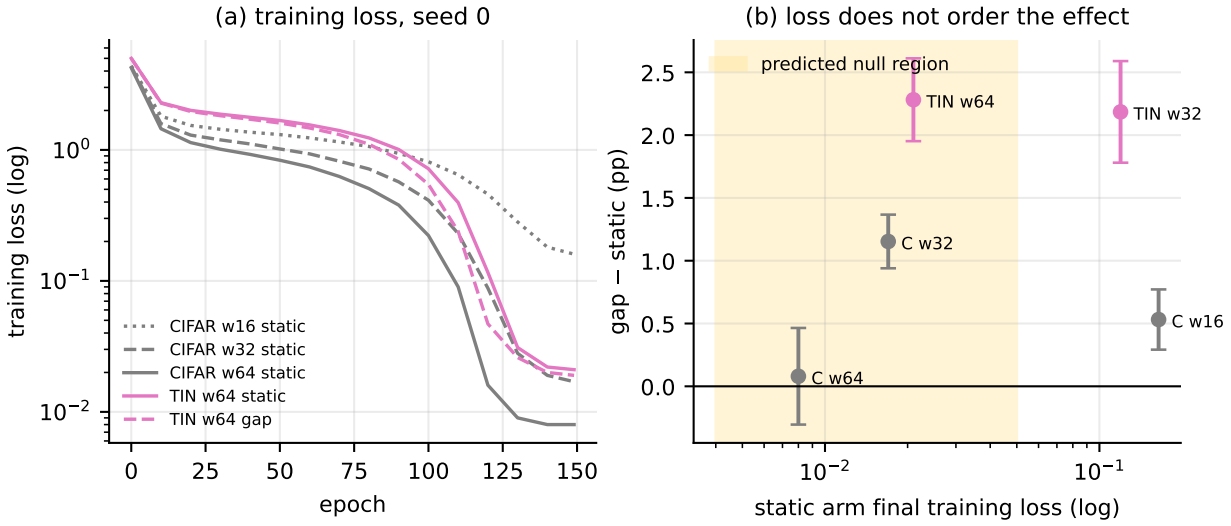


Figure 5: ResNet-18. (a) Training loss of static and gap arms, seed 0, log scale. (b) Routing effect against the final training loss of the static arm in all five conditions. The shaded band is the region the prediction associated with a null.

Table 4: ResNet-18: five conditions ordered by the final training loss of the static arm.

dataset	width	static train loss	`gap` − `static` (pp)	95 % CI	verdict
CIFAR-100	64	0.0080	+0.080	[-0.305, +0.465]	equivalent
CIFAR-100	32	0.0170	+1.153	[+0.940, +1.367]	higher
Tiny-ImageNet	64	0.0210	+2.282	[+1.952, +2.611]	higher
Tiny-ImageNet	32	0.1194	+2.185	[+1.781, +2.589]	higher
CIFAR-100	16	0.1643	+0.532	[+0.292, +0.771]	higher

6.2 Temperature: the null survives annealing

With $\tau = 1$ the bank could be undertrained, in which case the null would say nothing about dynamic convolution. Repeating `gap` with the annealed schedule leaves it in place: annealed `gap` against `static` is +0.032 pp with $p_{\text{TOST}} = 0.0025$, a tighter equivalence than the 0.0188 at $\tau = 1$ (Section 10).

What remains. On ResNet-18, on CIFAR-100, at width 64, routing has no effect; the result is well powered, holds under both schedules, and does not appear on Tiny-ImageNet or on MobileNetV2. Training loss does not explain it and temperature does not cause it. Plausible differences between the conditions include input resolution, final feature-map size, number of classes and the share of parameters in dynamic layers; none was isolated here, and the null remains unexplained.

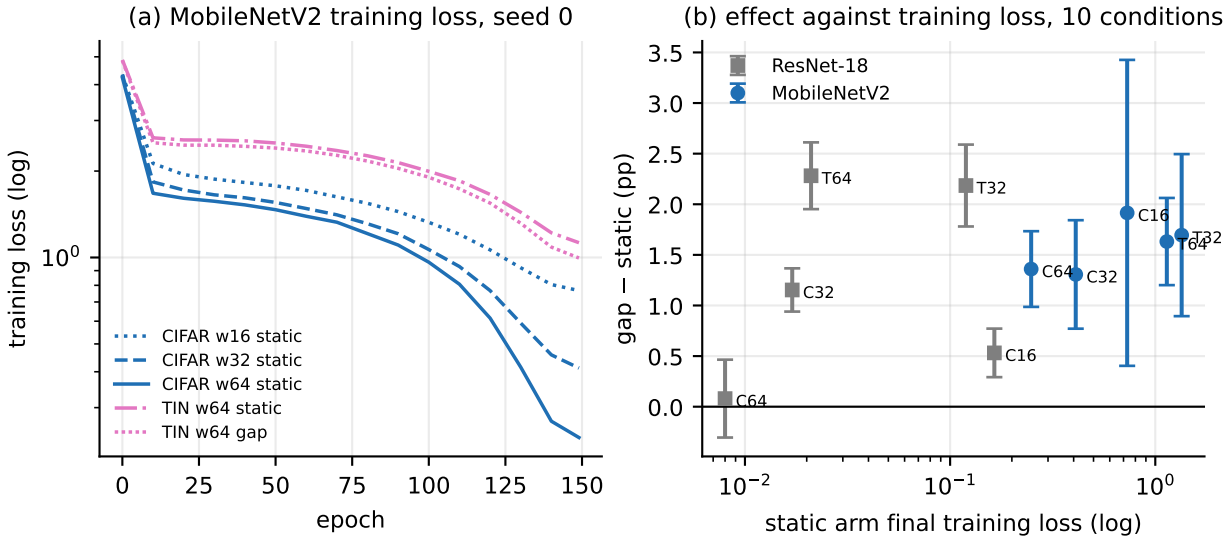


Figure 6: (a) MobileNetV2 training loss, seed 0, log scale. (b) `gap − static` against the static arm's final training loss for both backbones and all five conditions, 95 % CIs.

7 Does the router need to be trained?

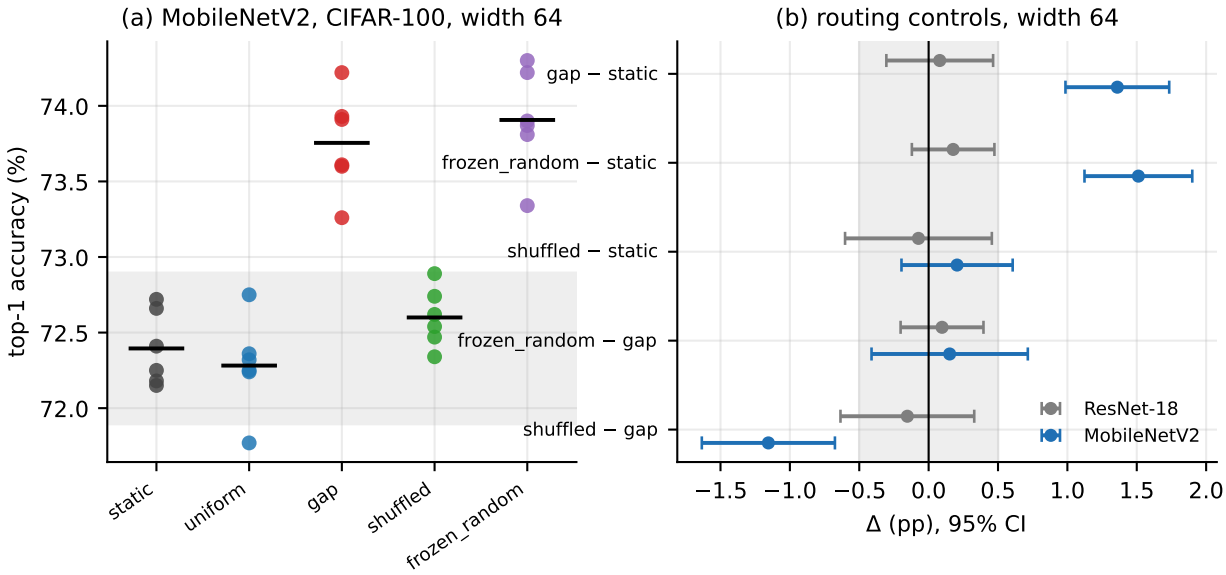


Figure 7: CIFAR-100, width 64. (a) Every MobileNetV2 run of the five arms; bars are arm means, the band is ±0.5 pp around `static`. (b) The routing controls on both backbones, 95 % CIs.

The controls `shuffled` and `frozen_random` were run on CIFAR-100 at width 64 (Table 5, Figure 7). They can only answer this question where routing has an effect to explain.

MobileNetV2. A router frozen at its random initialisation beats `static` by +1.512 pp and `uniform` by +1.625 pp ($p = 0.0008$), and is not distinguishable from the trained router (`frozen_random − gap` $= +0.152$ pp, 95 % CI [-0.411, +0.715], $p = 0.5196$). It is not equivalent to it at our margin either ($p_{\text{TOST}} = 0.0864$), so the claim is that no difference was detected, not that none exists. Routing each image with another image's descriptor removes the benefit: `shuffled` is not distinguishable from `static` (+0.205 pp, $p = 0.2456$) and falls below `gap` by -1.155 pp ($p = 0.0016$). Read together, the benefit on MobileNetV2 requires that each image's own descriptor determine its mixture, and does not require the router to be trained: a fixed random map from descriptors to mixtures is enough. The frozen router's weights are held as buffers and are not trained, so the gain is not a product of the 1137K trainable router parameters either.

ResNet-18. In the same condition every ResNet-18 arm, including both controls, is equivalent to the plain network, and `frozen_random − gap` is +0.097 pp ($p_{\text{TOST}} = 0.0089$). With no benefit to explain, this says nothing about whether routing needs to be trained. The finding is therefore MobileNetV2 only, and limited to CIFAR-100 at width 64.

What the routers output. The trained router is the furthest from uniform on both backbones (Table 6; the measurement is described in Section 3.5). On MobileNetV2 the frozen router is 4.5 times closer to uniform ($1 - \tilde{H} = 0.0029$ against 0.0132), but its departure is spread over the network: its entropy reaches 0.9999 in only 0–2 of the 17 dynamic layers. The `shuffled` router, whose input carries no information about the image it routes, trains itself to near-uniform on both backbones ($1 - \tilde{H} = 0.00005$ on MobileNetV2, 0.0004 on ResNet-18). On ResNet-18 the trained router's entropy is below that of the controls (Figure 3b), so it does vary with the input; at this width that variation changes nothing.

Table 5: Routing controls on CIFAR-100 at width 64, both backbones, paired by seed.

	ResNet-18		MobileNetV2				
comparison	Δ (pp)	verdict	Δ (pp)	95 % CI	p	p_{TOST}	verdict
`uniform − static`	+0.048	equivalent	-0.113	[-0.512, +0.285]	0.4978	0.0275	equivalent
`gap − static`	+0.080	equivalent	+1.360	[+0.986, +1.734]	0.0002	0.9990	higher
`shuffled − static`	-0.073	equivalent	+0.205	[-0.196, +0.606]	0.2456	0.0585	inconclusive
`frozen_random − static`	+0.177	equivalent	+1.512	[+1.123, +1.900]	0.0002	0.9994	higher
`frozen_random − gap`	+0.097	equivalent	+0.152	[-0.411, +0.715]	0.5196	0.0864	inconclusive
`shuffled − gap`	-0.153	inconclusive	-1.155	[-1.635, -0.675]	0.0016	0.9915	lower
`frozen_random − uniform`	+0.128	equivalent	+1.625	[+1.051, +2.199]	0.0008	0.9980	higher

Table 6: Routing entropy by arm, CIFAR-100 width 64, per-seed means over the recorded images and layers. Flat layers: the number of the 17 MobileNetV2 dynamic layers with $\tilde{H} \geq 0.9999$, range over seeds.

	ResNet-18		MobileNetV2				
arm	mean $\tilde{H}$	$1 - \tilde{H}$	mean $\tilde{H}$	min	max	$1 - \tilde{H}$	flat layers
`uniform`	1.0000	0.00000	1.00000	1.00000	1.00000	0.00000	17
`gap`	0.9885	0.01147	0.98680	0.98272	0.98955	0.01320	2–4
`frozen_random`	0.9971	0.00287	0.99706	0.99649	0.99736	0.00294	0–2
`shuffled`	0.9996	0.00040	0.99995	0.99992	0.99996	0.00005	14–16

8 Bank size

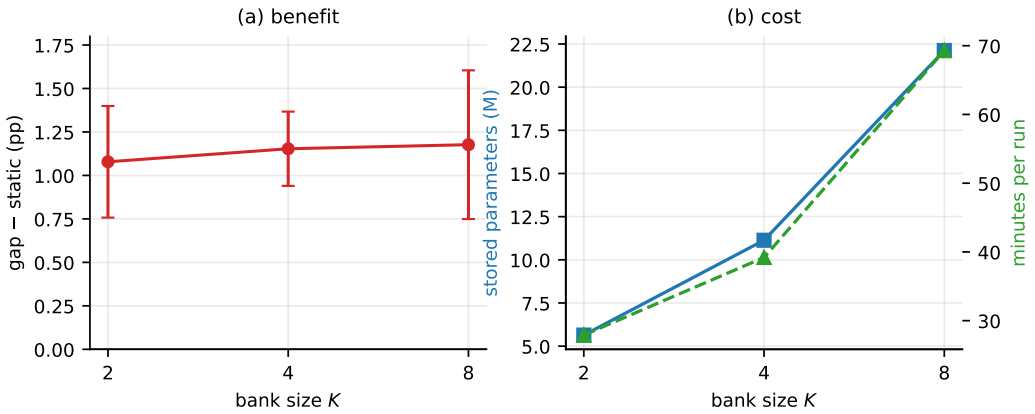


Figure 8: ResNet-18, CIFAR-100, width 32. (a) Effect over `static` for $K \in \{2, 4, 8\}$, 95 % CIs. (b) Stored parameters and training time.

At CIFAR-100 width 32 all three bank sizes beat `static` on both backbones (Tables 7 and 9, Figure 12b). On ResNet-18 no pair of sizes is distinguishable and every pair is equivalent (Table 8): $K = 8$ over $K = 2$ is +0.098 pp ($p = 0.409$). Going from $K = 2$ to $K = 8$ multiplies stored parameters by 3.9 and training time by 2.5 (Figure 8). On

MobileNetV2 $K = 8$ over $K = 2$ is +0.112 pp ($p = 0.5929$), not distinguishable but not equivalent at our margin either. On both backbones, then, nothing beyond $K = 2$ was detected.

Table 7: ResNet-18: bank size at CIFAR-100 width 32. Differences are against the `static` row.

K	accuracy	sd	Δ (pp)	95 % CI	p	params (M)	min/run
2	75.950	0.148	+1.078	[+0.758, +1.399]	0.0003	5.64	27.9
4	76.025	0.156	+1.153	[+0.940, +1.367]	$< 10^{-4}$	11.13	39.2
8	76.048	0.242	+1.177	[+0.749, +1.604]	0.0009	22.12	69.3
`static`	74.872	0.262	–	–	–	2.82	8.6

Table 8: ResNet-18: pairwise comparisons between bank sizes, CIFAR-100 width 32.

comparison	Δ (pp)	95 % CI	p	p_{TOST}	verdict
$K = 4$ vs $K = 2$	+0.075	[-0.087, +0.237]	0.2878	0.0005	equivalent
$K = 8$ vs $K = 4$	+0.023	[-0.286, +0.332]	0.8538	0.0053	equivalent
$K = 8$ vs $K = 2$	+0.098	[-0.182, +0.379]	0.4091	0.0072	equivalent

Table 9: MobileNetV2: bank size at CIFAR-100 width 32.

K	accuracy	sd	Δ (pp)	95 % CI	p	params (M)	min/run
2	71.773	0.490	+0.998	[+0.487, +1.510]	0.0041	1.14	51.6
4	72.082	0.356	+1.307	[+0.771, +1.843]	0.0015	1.20	47.3
8	71.885	0.400	+1.110	[+0.647, +1.573]	0.0016	1.34	98.8
`static`	70.775	0.217	–	–	–	0.82	18.0

9 Is routing worth its cost?

The answer depends on what is held fixed.

9.1 Equal parameters

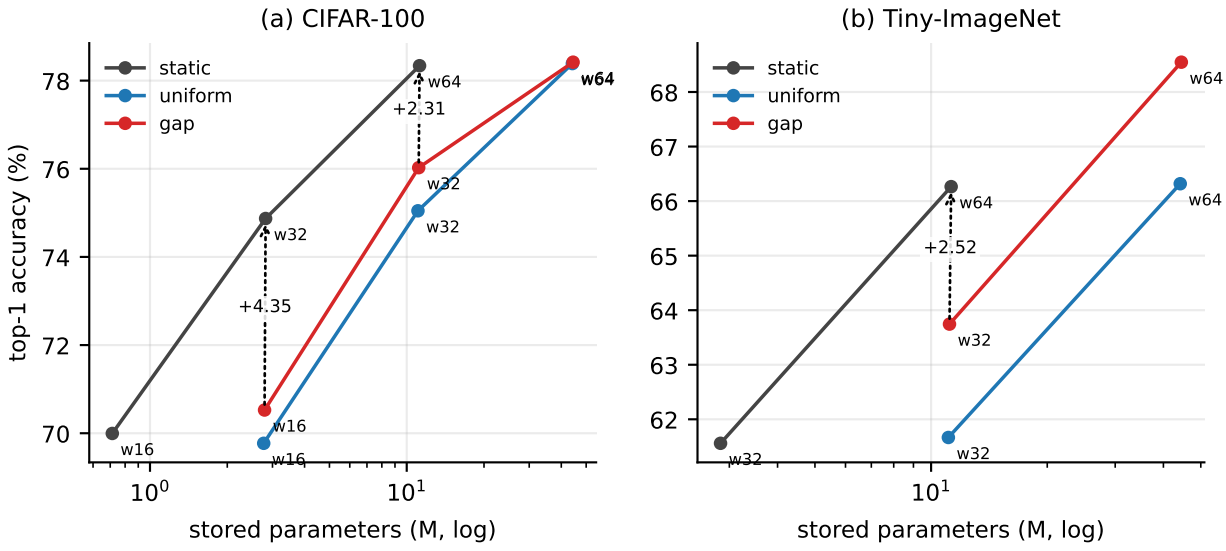


Figure 9: ResNet-18: accuracy against stored parameters. Dotted arrows join routed and plain networks matched to within 1 % on parameter count.

On ResNet-18 the widths differ by a factor of two and the bank multiplies convolution parameters by four, so a routed network at one width stores almost exactly as many parameters as a plain network at the next width up. That gives three matched comparisons without any extra runs (Table 10, Figure 9). The plain network wins all three: by +4.345 pp at 2.8M parameters and +2.313 pp at 11.2M on CIFAR-100, and by +2.520 pp at 11.2M on Tiny-ImageNet. It also trains faster in each case; on Tiny-ImageNet, 129.0 min per run against 284.2 min. A fourth, unmatched comparison points the same way: the $K = 8$ network at width 32 stores 22.1M parameters, twice the plain width-64 network, and trails it by 2.29 pp.

On MobileNetV2 the same pairs are not matched: so few parameters sit in dynamic layers that the routed network has only 0.60 to 0.51 times the parameters of the plain network at twice its width (Table 11, Figure 10). Even so, the plain network wins only at the smallest width (+2.503 pp, $p = 0.0001$). On CIFAR-100 at width 32 and on Tiny-ImageNet the two are not distinguishable (+0.313 pp, $p = 0.1111$; +0.155 pp, $p = 0.5806$). Interpolating the static accuracy curve in log-parameters to the routed network's parameter count puts the routed network ahead in all three cases (static minus routed: -0.298, -0.713 and -1.037 pp); that comparison is an interpolation and has no confidence interval.

Table 10: ResNet-18: routed against plain networks at matched parameter counts, paired by seed. Minutes are routed / plain per run. The last column is the routed network's stored parameters divided by the plain one's; a pair is called matched only within 10 %.

dataset	routed	plain	routed	plain	Δ (pp)	95 % CI	p	min/run	param ratio
CIFAR-100	`gap` w16 (2.79M)	`static` w32 (2.82M)	70.527	74.872	+4.345	[+4.103, +4.587]	$< 10^{-4}$	33.7 / 8.6	0.99 (matched)
CIFAR-100	`gap` w32 (11.13M)	`static` w64 (11.22M)	76.025	78.338	+2.313	[+1.870, +2.757]	$< 10^{-4}$	39.2 / 18.8	0.99 (matched)
Tiny-ImageNet	`gap` w32 (11.16M)	`static` w64 (11.27M)	63.745	66.265	+2.520	[+2.311, +2.729]	$< 10^{-4}$	284.2 / 129.0	0.99 (matched)

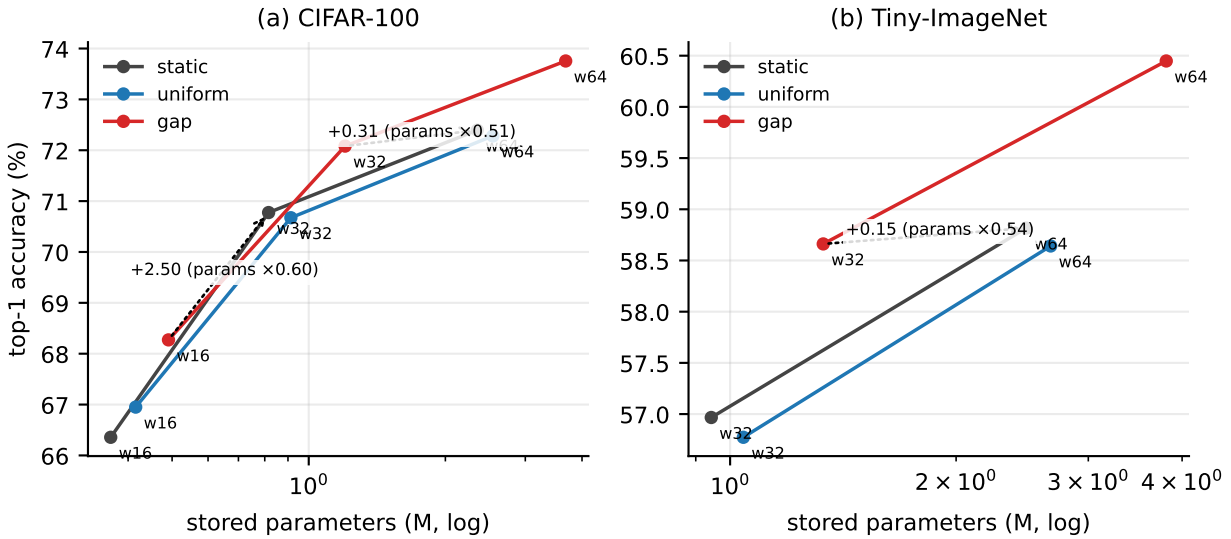


Figure 10: MobileNetV2: accuracy against stored parameters. Arrows join `gap` at width c to `static` at width $2c$; labels give the accuracy difference and the parameter ratio.

Table 11: MobileNetV2: `gap` at width c against `static` at width $2c$, paired by seed. Δ is plain minus routed. The ratio is the routed network's stored parameters over the plain one's; none of these pairs is matched. The last column compares `gap` with the static curve interpolated to its parameter count.

dataset	routed	plain	routed	plain	Δ (pp)	95 % CI	p	verdict	ratio	Δ equal params
CIFAR-100	`gap` w16 (0.49M)	`static` w32 (0.82M)	68.272	70.775	+2.503	[+1.927, +3.080]	0.0001	higher	0.60	-0.298
CIFAR-100	`gap` w32 (1.20M)	`static` w64 (2.35M)	72.082	72.395	+0.313	[-0.103, +0.730]	0.1111	inconclusive	0.51	-0.713
Tiny-ImageNet	`gap` w32 (1.33M)	`static` w64 (2.48M)	58.662	58.817	+0.155	[-0.520, +0.830]	0.5806	inconclusive	0.54	-1.037

9.2 Equal compute

Dynamic convolution is usually advertised at equal compute, and there the picture is different (Table 12). Doubling the width costs three to four times the multiply-accumulates of the routed network; routing at the same width costs at most 8 % more on ResNet-18 and 2 % more on MobileNetV2, counted for an implementation that aggregates the kernel per sample. At the same width, routing beats the plain network in 4 of five conditions on ResNet-18 and 5 of five on

MobileNetV2. Our training implementation computes K convolutions per dynamic layer and is slower in wall-clock time than either count suggests (Tables 21 and 27); that is a property of the implementation, not of the layer.

Summary. At equal parameters, width wins on ResNet-18 in all three pairs and on MobileNetV2 in one of three, against a plain network with 1.7 to 2.0 times the parameters. At equal width, routing wins on both backbones wherever it has an effect. Whether dynamic convolution is worth its cost therefore depends on the cost held fixed and on the backbone.

Table 12: The routed network under two cost views. Left: `static` at width $2c$ minus `gap` at width c, with the parameter ratio routed/plain and the multiply-accumulate (MAC) ratio plain/routed. Right: `gap` minus `static` at the same width, with the MAC ratio routed/plain. MACs per image are counted for per-sample kernel aggregation.

		plain at $2c$ vs routed at c			routed vs plain, same width c	
backbone	condition	param ratio	MAC ratio	plain − routed (pp)	MAC ratio	routed − plain (pp)
ResNet-18	CIFAR-100 w16	0.99	3.7×	+4.35 (higher)	1.08×	+0.53 (higher)
ResNet-18	CIFAR-100 w32	0.99	3.7×	+2.31 (higher)	1.08×	+1.15 (higher)
ResNet-18	Tiny-ImageNet w32	0.99	3.9×	+2.52 (higher)	1.02×	+2.18 (higher)
MobileNetV2	CIFAR-100 w16	0.60	2.9×	+2.50 (higher)	1.01×	+1.91 (higher)
MobileNetV2	CIFAR-100 w32	0.51	3.1×	+0.31 (inconclusive)	1.01×	+1.31 (higher)
MobileNetV2	Tiny-ImageNet w32	0.54	3.1×	+0.15 (inconclusive)	1.00×	+1.70 (higher)

10 Temperature annealing

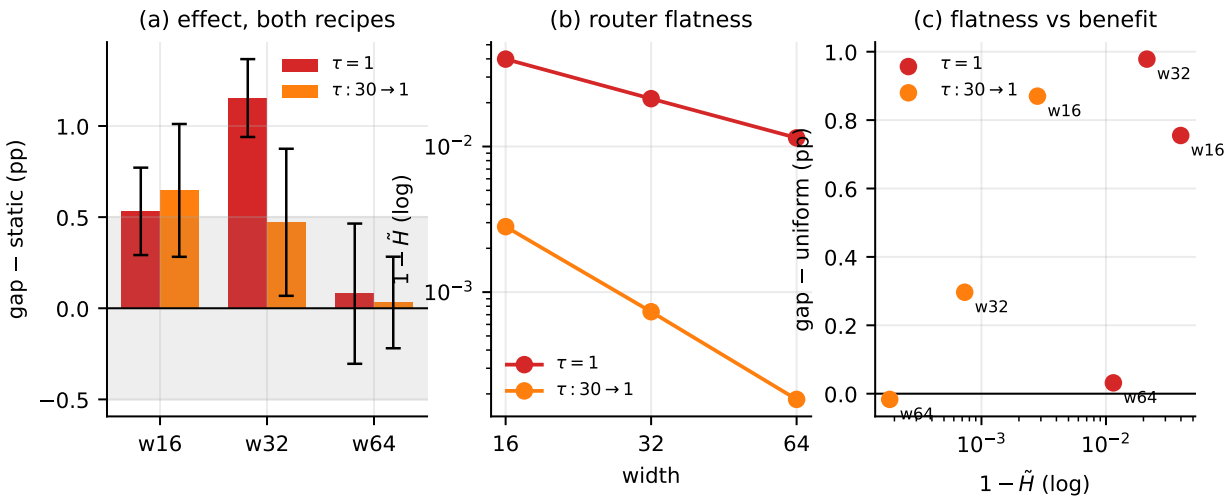


Figure 11: ResNet-18, CIFAR-100. (a) Routing effect under a constant and an annealed temperature, 95 % CIs. (b) Router flatness. (c) Flatness against benefit over `uniform`; the two schedules do not lie on one curve.

With $\tau = 1$ the softmax can be close to one-hot from the first step, starving most of the bank of gradient. DY-Conv anneals τ from 30 to 1 for this reason (Chen et al., 2020). We repeated `gap` at all three CIFAR-100 widths with that schedule on both backbones (Tables 13 and 14, Figures 11 and 12a).

Accuracy. Annealing never raised accuracy. On MobileNetV2 it lowered `gap` at every width: -1.160 pp at width 16 ($p = 0.0352$), -1.277 pp at width 32 ($p = 0.0014$) and -0.922 pp at width 64 ($p = 0.0011$), and the annealed runs ended at a higher training loss at every width (0.713, 0.411, 0.225 against 0.655, 0.344, 0.168). On ResNet-18 it lowered accuracy at width 32 (-0.682 pp, $p = 0.0017$, again with a higher final training loss, 0.0173 against 0.0132) and made no detectable difference at width 16 (+0.115 pp, $p = 0.4313$) or width 64 (-0.048 pp, $p = 0.7449$). Over the six conditions annealing is lower in 4 and higher in 0. Backbone, dataset, bank size and schedule length all differ from the DY-Conv setting, so this is not evidence against their result.

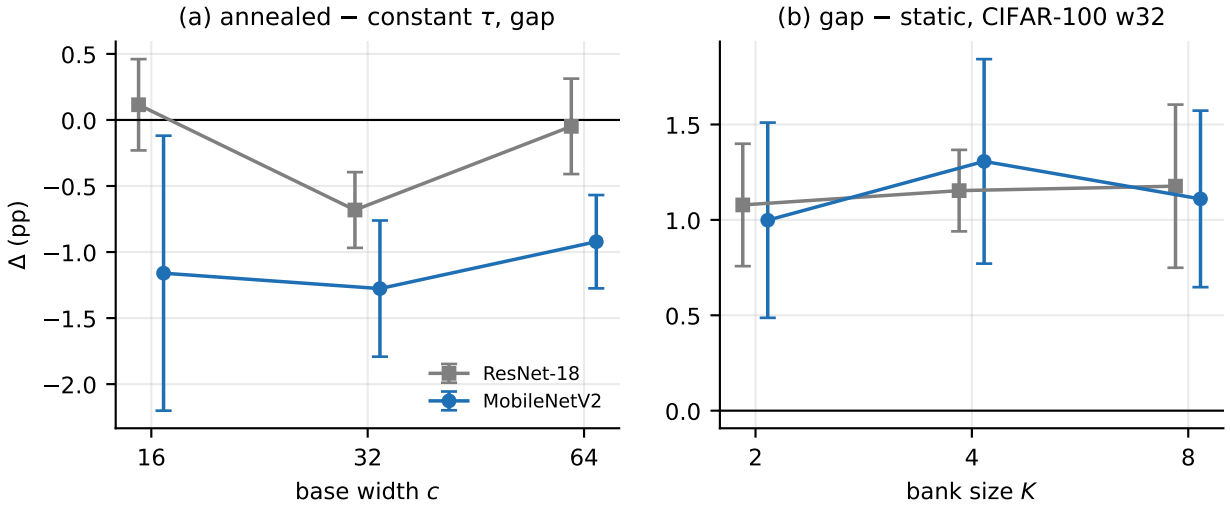


Figure 12: Both backbones, CIFAR-100. (a) Annealed minus constant temperature for `gap`. (b) `gap` − `static` at width 32 for $K \in \{2, 4, 8\}$. 95 % CIs.

Annealed routers against the constant mixture. Compared with `uniform`, the annealed router is equivalent to it on ResNet-18 at width 64 (-0.017 pp, $p_{\text{TOST}} = 0.0007$) and on MobileNetV2 at width 32 (+0.132 pp); it is inconclusive on ResNet-18 at width 32 (+0.297 pp) and on MobileNetV2 at width 16 (+0.162 pp); and it remains above `uniform` on ResNet-18 at width 16 (+0.870 pp, $p = 0.0044$) and on MobileNetV2 at width 64 (+0.552 pp). After annealing the routers are close to uniform by entropy (Section 11). One reading follows from Proposition 1: a nearly constant mixture trains every kernel toward a common update and so makes the bank homogeneous, which leaves a router little to choose between. We did not test this, and entropy was recorded only at the end of training, so we cannot say when any router became flat.

Table 13: ResNet-18: temperature ablation on CIFAR-100, `gap` arm. The last two columns test whether the annealed router behaves like the constant mixture.

	accuracy		annealed − constant		vs `static`			annealed vs `uniform`	
width	$\tau = 1$	annealed	Δ	p	$\tau = 1$	annealed	p_{TOST}	Δ	verdict
16	70.527	70.642	+0.115	0.4313	+0.532	+0.647	0.8259	+0.870	higher
32	76.025	75.343	-0.682	0.0017	+1.153	+0.472	0.4319	+0.297	inconclusive
64	78.418	78.370	-0.048	0.7449	+0.080	+0.032	0.0025	-0.017	equivalent

Table 14: MobileNetV2: temperature ablation on CIFAR-100, `gap` arm.

	accuracy		annealed − constant		vs `static`			annealed vs `uniform`	
width	$\tau = 1$	annealed	Δ	p	$\tau = 1$	annealed	p	Δ	verdict
16	68.272	67.112	-1.160	0.0352	+1.915	+0.755	0.3286	+0.162	inconclusive
32	72.082	70.805	-1.277	0.0014	+1.307	+0.030	0.8534	+0.132	equivalent
64	73.755	72.833	-0.922	0.0011	+1.360	+0.438	0.0051	+0.552	higher

11 Routing entropy

Routing entropy (Section 3.5) is the statistic this literature reports to show that a router is doing something. Tables 15 and 16 set it against the benefit each router delivers over `uniform`. Entropy compresses the signal quadratically. Writing $z_k = 1/K + \varepsilon_k$ with $\sum_k \varepsilon_k = 0$ and expanding to second order,

$$1 - \tilde{H} \;\approx\; \frac{K}{2 \log K} \sum_k \varepsilon_k^2, \tag{4}$$

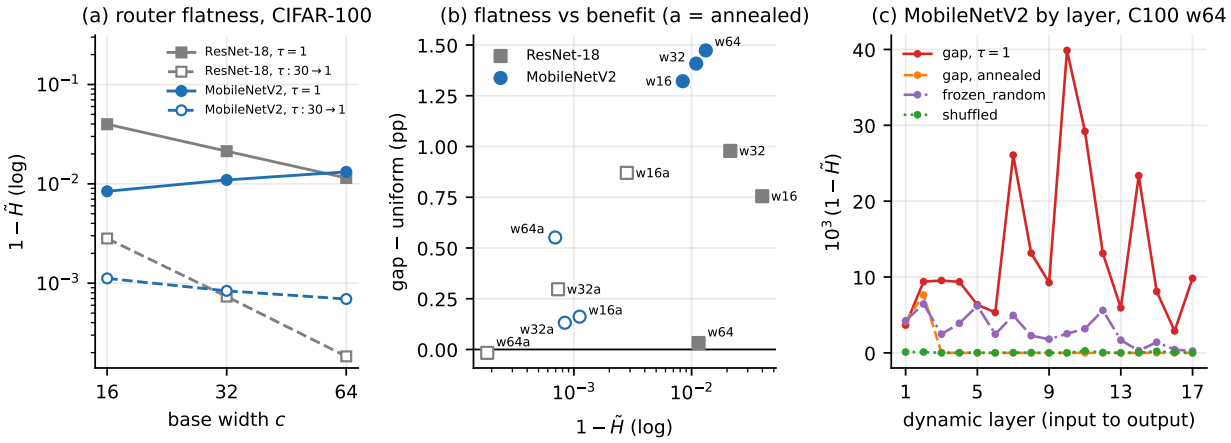


Figure 13: Routing entropy on both backbones. (a) Departure from uniform, $1 - \tilde{H}$, against width on CIFAR-100; open markers and dashed lines are the annealed schedule. (b) $1 - \tilde{H}$ against the benefit over `uniform` in the six CIFAR-100 conditions of each backbone; "a" marks the annealed schedule. (c) MobileNetV2, CIFAR-100 width 64: $1 - \tilde{H}$ for each dynamic layer, mean over six seeds.

so the deviations that carry input-dependent information enter squared. At $1 - \tilde{H} = 0.00282$, for example, the root-mean-square deviation from $1/K$ is about 0.022, so typical mixing weights lie in [0.228, 0.272], roughly 9 % of $1/K$, and entropy reports that as 0.997. We call a layer *flat* when $\tilde{H} \geq 0.9999$, which allows a root-mean-square deviation from $1/K$ of about 0.004, under 2 % of $1/K$.

Within ResNet-18, entropy does not order the benefit. Across its six CIFAR-100 conditions the rank correlation between $1 - \tilde{H}$ and the benefit over `uniform` is 0.60 (Spearman, $p = 0.208$). At width 16 the two schedules give a counterexample (Figure 14): the constant router has $1 - \tilde{H} = 0.03983$ and beats `uniform` by +0.755 pp, while the annealed router is 14 times closer to uniform ($1 - \tilde{H} = 0.00282$) and beats it by +0.870 pp ($p = 0.0044$).

Within MobileNetV2, it does. The same correlation is 0.83 ($p = 0.042$), and the width-16 counterexample does not recur: the annealed router is 7.5 times closer to uniform and delivers less (+0.16 pp, $p = 0.7242$, against +1.32 pp). Inside the annealed schedule the order is reversed: the router closest to uniform, at width 64, is the only annealed one whose benefit is significant (+0.55 pp, $p = 0.0133$, against +0.16 and +0.13 pp at widths 16 and 32); the differences between these three benefits were not tested.

Across backbones, the same entropy, a different benefit. On CIFAR-100 at width 64 the constant-temperature routers of the two backbones are almost equally far from uniform ($1 - \tilde{H} = 0.0115$ on ResNet-18, 0.0132 on MobileNetV2). The ResNet-18 router is equivalent to `uniform` (+0.03 pp); the MobileNetV2 router beats it by +1.47 pp ($p < 10^{-4}$). The width trends are opposite: from width 16 to 64 the ResNet-18 router moves towards uniform (0.0398 to 0.0115) and the MobileNetV2 router away from it (0.0084 to 0.0132; Figure 13a).

After annealing, routers end close to uniform. Under the annealed schedule the MobileNetV2 router is flat in 15 to 16 of its 17 dynamic layers, in every seed at every width; across all eighteen annealed runs the only layers that are ever not flat are layers 1, 2 and 4, counted from the input (Figure 13c, Table 26). Under a constant temperature 2 to 7 layers are flat. The temperature reaches 1 at epoch 10, so these routers end close to uniform after 140 epochs at the same temperature as the constant ones. On ResNet-18 the annealed router at width 64 ends at $1 - \tilde{H} = 0.00018$ and is equivalent to `uniform`; on MobileNetV2 the annealed router at width 64 ends at $1 - \tilde{H} = 0.00069$ and still beats `uniform` by +0.55 pp. A very low departure from uniform thus accompanied no benefit in one case and a significant benefit in the other.

Conclusion. Entropy can move with the benefit, as within MobileNetV2, but it did not within ResNet-18, and the same value accompanies very different benefits on the two backbones. A value of $\tilde{H}$ below one is not evidence that routing is useful, and a value near one is not evidence that it is not. These statements rest on 16 images per run, measured once at the end of training (Section 3.5).

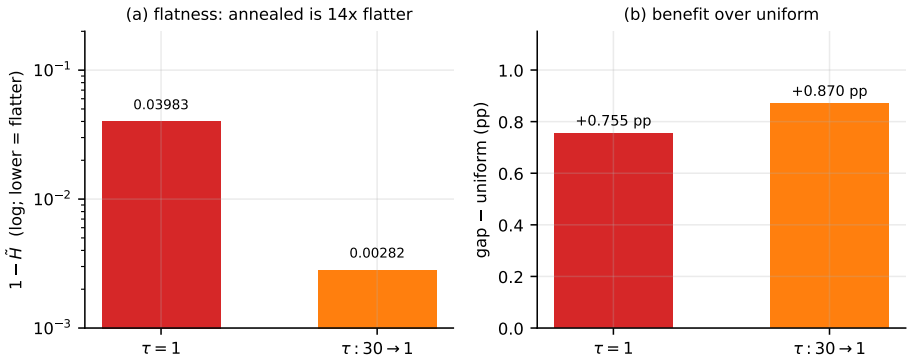


Figure 14: ResNet-18, CIFAR-100, width 16. The annealed router is 14 times closer to uniform by entropy and delivers the larger benefit over `uniform`. On MobileNetV2 the order reverses (Table 16).

Table 15: ResNet-18: routing entropy against the benefit it is taken to indicate, CIFAR-100. RMS dev is the root-mean-square deviation of the mixing weights from $1/K$, recovered from Eq. (4).

width	τ	$\tilde{H}$	$1-\tilde{H}$	RMS dev	`gap−uniform` (pp)	p
16	1	0.96017	0.03983	0.0831	+0.755	0.0023
16	30→1	0.99718	0.00282	0.0221	+0.870	0.0044
32	1	0.97868	0.02132	0.0608	+0.978	0.0010
32	30→1	0.99927	0.00073	0.0113	+0.297	0.0810
64	1	0.98853	0.01147	0.0446	+0.032	0.7752
64	30→1	0.99982	0.00018	0.0056	-0.017	0.8369

Table 16: MobileNetV2: routing entropy against the benefit over `uniform`, for every `gap` group that has a `uniform` arm. RMS dev is recovered from Eq. (4) as in Table 15. Flat layers: dynamic layers (of 17) with $\tilde{H} \geq 0.9999$, range over seeds.

dataset	width	τ	$\tilde{H}$	$1-\tilde{H}$	RMS dev	flat layers	`gap−uniform` (pp)	p	verdict
CIFAR-100	16	1	0.99159	0.00841	0.0382	6–7	+1.322	0.0011	higher
CIFAR-100	16	30→1	0.99888	0.00112	0.0139	15–16	+0.162	0.7242	inconclusive
CIFAR-100	32	1	0.98906	0.01094	0.0435	4–7	+1.408	0.0002	higher
CIFAR-100	32	30→1	0.99916	0.00084	0.0120	15	+0.132	0.4319	equivalent
CIFAR-100	64	1	0.98680	0.01320	0.0478	2–4	+1.473	$< 10^{-4}$	higher
CIFAR-100	64	30→1	0.99931	0.00069	0.0110	15–16	+0.552	0.0133	higher
Tiny-ImageNet	32	1	0.99267	0.00733	0.0356	3–5	+1.888	0.0008	higher
Tiny-ImageNet	64	1	0.99160	0.00840	0.0382	2–5	+1.807	0.0014	higher

12 The pre-registered predictions

Four predictions were recorded, dated, before the runs that test them, and three failed. The ResNet-18 prediction that the effect would vanish on Tiny-ImageNet at width 64 failed (Table 3, Section 6). Table 17 gives the three MobileNetV2 predictions, recorded on 30 September 2026. P1 failed: the bank term was equivalent to zero in 2 of five conditions. The three inconclusive conditions are not evidence of a bank effect, since none is significant, and finding 1 claims only that no bank effect was detected; but the prediction was equivalence, and we score it as stated. P2 failed, holding in 1 of three pairs, a reversal of the ResNet-18 result (Section 9). P3 held in 5 of five conditions.

Table 17: MobileNetV2: predictions recorded on 30 September 2026, before any MobileNetV2 run, and the outcomes.

	prediction	measured	outcome
P1	The bank term (`uniform−static`) is equivalent to zero (TOST, ± 0.5 pp) in each of the five conditions	2 of 5	failed
P2	`static` at width $2c$ is more accurate than `gap` at width c in each of the three pairs	1 of 3	failed
P3	The routing term is positive with $p < 0.05$ in at least three of the five conditions	5 of 5	held

13 Result by result

Table 18 lists the main results on both backbones. Of its 10 rows, 4 hold on both backbones, 4 depend on the backbone, one (the frozen router) is informative on MobileNetV2 only, and one compares the backbones directly.

Table 18: The main results on both backbones. Differences are in pp and paired by seed. Status: *holds on both*, *backbone-dependent*, *MobileNetV2 only* (uninformative on ResNet-18) or *across backbones* (a comparison between them).

result	ResNet-18	MobileNetV2	status
Bank term (`uniform − static`)	equivalent in 5 of 5	equivalent in 2 of 5; never significant (largest +0.59)	holds on both (weaker on MobileNetV2)
Routing term significant	4 of 5	5 of 5	holds on both
`gap − static`, CIFAR-100 w64	+0.08 (equivalent)	+1.36 (higher)	**backbone-dependent**
Plain $2c$ beats `gap` c (3 pairs)	3 of 3 (+4.35, +2.31, +2.52)	1 of 3 (+2.50, +0.31, +0.15)	**backbone-dependent**
Annealed − constant τ (w16, 32, 64)	+0.11, -0.68, -0.05	-1.16, -1.28, -0.92	holds on both (never higher)
`frozen_random − gap`, w64	+0.10 (both null)	+0.15; vs `uniform` +1.63	MobileNetV2 only
$K = 8$ vs $K = 2$, CIFAR-100 w32	+0.10 (equivalent)	+0.11 (inconclusive)	holds on both
Entropy counterexample, CIFAR-100 w16	annealed 14× flatter, larger benefit (+0.87 vs +0.76)	annealed 7.5× flatter, smaller benefit (+0.16 vs +1.32)	**backbone-dependent**
Constant-τ $1 - \tilde{H}$ as width grows (w16, 32, 64)	0.0398, 0.0213, 0.0115 (flattens)	0.0084, 0.0109, 0.0132 (sharpens)	**backbone-dependent**
Same $1 - \tilde{H}$, CIFAR-100 w64, `gap−uniform`	0.0115: +0.03 (equivalent)	0.0132: +1.47 (higher)	across backbones

An interpretation. On MobileNetV2 the router does not need to be trained, and annealing, which keeps the mixture nearly constant early in training and leaves it close to uniform at the end, removes most of the benefit. Both fit the reading that what matters is that the mixture differ across images from early in training, not that it be optimised. ResNet-18 does not support this reading everywhere: at width 16 the annealed router kept the full benefit over `uniform` (+0.870 against +0.755 pp). Neither experiment was designed to test it, and we state it as an interpretation only.

Why the backbones may differ. Several properties change at once between ResNet-18 and MobileNetV2: the share of parameters in dynamic layers (nearly all against 3 %), full against depthwise convolution in those layers, the size of the router relative to the bank, and how close the network comes to fitting its training set under this recipe. None was isolated here, so none of the backbone-dependent results can be attributed to one of them. The three remaining backbones vary these properties: VGG-16 keeps every parameter in full dynamic convolutions, ResNet-50 about half, and ConvNeXt-Tiny makes only its depthwise 7×7 layers dynamic.

14 Threats to validity

Scale. Both datasets are low-resolution and every run fits on one consumer GPU. Behaviour at ImageNet scale is untested and cannot be tested on this hardware.

Two backbones. The backbones disagree on 4 of the 10 results in Table 18, so what holds on both is the minimum that can be claimed. The same program on VGG-16, ResNet-50 and ConvNeXt-Tiny was pre-registered on 30 September 2026 and is running; its results will be added as they complete, whatever they show.

Placement and router size on MobileNetV2. Copying the ResNet-18 rule makes only the depthwise layers of MobileNetV2 dynamic, 3 % of its parameters, and its router is larger than the bank. A different rule, such as making every convolution but the stem dynamic, could change the MobileNetV2 results. The frozen-router control shows that the trainable router parameters do not produce the gain.

Entropy sample. The routing statistics are overwritten at each test batch, so every recorded entropy, on both backbones, describes the last batch of the unshuffled test set: the same 16 images in every run on a dataset. Comparisons between arms, widths and schedules are therefore made on identical inputs, and the spread across seeds is small (Tables 22 and 6), but no entropy here is a test-set average. The trained weights were not kept, so the entropies cannot be recomputed on the full test set. No accuracy is affected.

One recipe. Both backbones use the same recipe. The MobileNetV2 static arms never approach zero training loss under it, and a recipe tuned for MobileNetV2 could change where routing helps.

Initialisation. On ResNet-18, static convolutions in the `static` arm, and the static stem, shortcut and head layers in every arm, keep PyTorch's default initialisation (Kaiming-uniform with $a = \sqrt{5}$), while every kernel of every dynamic bank is initialised Kaiming-normal with fan-out. The ResNet-18 bank term therefore compares two initialisations as well as a bank against a single kernel. The routing term is not affected: both arms have identically initialised banks. MobileNetV2, and the replication on three more backbones, initialise static and bank kernels with one scheme, and the MobileNetV2 bank term, which is free of the mismatch, also shows no bank effect.

Not a reproduction of a published system. The layer is a dynamic convolution of the CondConv and DY-Conv form. No published implementation of ODConv, FDConv or KernelWarehouse was run, and the conclusions concern dynamic convolution of this form.

Global routing. Every router here uses a global descriptor. Spatially resolved routing is not covered.

One annealing schedule. Section 10 compares a constant temperature with one linear schedule.

Stored and effective parameters. By Proposition 1 the `uniform` arm has the effective capacity of a plain network. The equal-parameter comparisons of Section 9.1 use stored parameters; the equal-compute comparison of Section 9 favours routing on both backbones.

The ResNet-18 CIFAR-100 width-64 null is unexplained. It is well powered and does not depend on the temperature schedule, its stated explanation failed a recorded test, and it does not appear on MobileNetV2 (Section 6).

Provenance and corrections. ResNet-18 final accuracies were transcribed from the training logs, which print them verbatim, and every figure and table is generated from those per-seed values; MobileNetV2 values are parsed from its log and checked against its result files (Section 3). During the ResNet-18 analysis, three scripts grouped results by width alone, which merged runs from the two datasets and from different bank sizes; this was detected when a comparison paired one dataset's `uniform` arm with the other's `static` arm, and every number in this report was recomputed from per-seed values keyed on dataset, width, bank size and schedule. Summary sentences printed by the analysis scripts were also found to over-read non-significant orderings on four occasions; this report relies on the per-comparison statistics only. The report generator was later rewritten to read the result files directly; on the 24 ResNet-18 width-64 result files checked it reproduces the transcribed accuracies and routing entropies exactly. An earlier version of this report printed stored parameter counts with a third decimal that had not been measured, and six of nineteen were off by up to 0.005 M (for example 0.708 M for 0.713 M); the counts here are counted from the model definition and agree with the training logs to the two decimals they print. No comparison changes.

15 Conclusion

Across two backbones, two datasets and 264 runs, the gain of dynamic convolution does not come from its kernel bank. A uniformly mixed bank was never significantly better or worse than a plain network, and was equivalent to one in 7 of ten conditions. Wherever dynamic convolution beat the plain network, which happened in 9 of ten conditions, input-dependent mixing carried the effect, and banks larger than $K = 2$ added nothing that we could detect. On MobileNetV2 that mixing did not have to be learned: a router frozen at random initialisation did as well as a trained one, provided each image was routed by its own descriptor.

How much this is worth depends on the backbone and on the cost held fixed. At equal parameters, a plain ResNet-18 of twice the width did better by 2.3 to 4.3 pp, while on MobileNetV2 a plain network with 1.7 to 2.0 times the parameters won only one of three pairs. At equal width, for a few percent more compute, routing won on both backbones wherever it had an effect. The one condition without an effect, ResNet-18 on CIFAR-100 at width 64, remains unexplained.

Two common practices did not help. Annealing the softmax temperature never raised accuracy and lowered it in 4 of six conditions, leaving the routers close to uniform. Routing entropy did not reliably indicate whether routing helped: it ordered the benefit within MobileNetV2 but not within ResNet-18, and the same entropy accompanied benefits of +0.03 and +1.47 pp on the two backbones. Three of our four recorded predictions failed.

Reproducibility. 264 runs (132 ResNet-18, 132 MobileNetV2), 458 GPU-hours, one RTX 4070 Ti. Code, the pre-registrations, the dated prediction, the gate ledgers with recorded evidence, every result file and every per-seed value accompany this report.

References

S. Arora, N. Cohen, and E. Hazan. On the optimization of deep networks: Implicit acceleration by overparameterization. In *ICML*, 2018.

J. Chen, X. Wang, Z. Guo, X. Zhang, and J. Sun. Dynamic region-aware convolution. In *CVPR*, 2021.

L. Chen, L. Gu, L. Li, C. Yan, and Y. Fu. Frequency dynamic convolution for dense image prediction. In *CVPR*, pages 30178–30188, 2025.

Y. Chen, X. Dai, M. Liu, D. Chen, L. Yuan, and Z. Liu. Dynamic convolution: Attention over convolution kernels. In *CVPR*, pages 11030–11039, 2020.

X. Ding, Y. Guo, G. Ding, and J. Han. ACNet: Strengthening the kernel skeletons for powerful CNN via asymmetric convolution blocks. In *ICCV*, 2019.

X. Ding, X. Zhang, N. Ma, J. Han, G. Ding, and J. Sun. RepVGG: Making VGG-style ConvNets great again. In *CVPR*, 2021.

S. He, C. Jiang, D. Dong, and L. Ding. SD-Conv: Towards the parameter-efficiency of dynamic convolution. In *WACV*, 2023.

K. He, X. Zhang, S. Ren, and J. Sun. Deep residual learning for image recognition. In *CVPR*, 2016.

A. Krizhevsky. Learning multiple layers of features from tiny images. Technical report, University of Toronto, 2009.

D. Lakens. Equivalence tests: A practical primer for t tests, correlations, and meta-analyses. *Social Psychological and Personality Science*, 8(4):355–362, 2017.

Y. Le and X. Yang. Tiny ImageNet visual recognition challenge. CS231N course report, Stanford University, 2015.

C. Li and A. Yao. KernelWarehouse: Rethinking the design of dynamic convolution. In *ICML*, PMLR 235:29201–29221, 2024.

C. Li, A. Zhou, and A. Yao. Omni-dimensional dynamic convolution. In *ICLR*, 2022.

Y. Li, Y. Chen, X. Dai, M. Liu, D. Chen, Y. Yu, L. Yuan, Z. Liu, M. Chen, and N. Vasconcelos. Revisiting dynamic convolution via matrix decomposition. In *ICLR*, 2021.

M. Sandler, A. Howard, M. Zhu, A. Zhmoginov, and L.-C. Chen. MobileNetV2: Inverted residuals and linear bottlenecks. In *CVPR*, 2018.

D. J. Schuirmann. A comparison of the two one-sided tests procedure and the power approach for assessing the equivalence of average bioavailability. *Journal of Pharmacokinetics and Biopharmaceutics*, 15(6):657–680, 1987.

F. Wilcoxon. Individual comparisons by ranking methods. *Biometrics Bulletin*, 1(6):80–83, 1945.

B. Yang, G. Bender, Q. V. Le, and J. Ngiam. CondConv: Conditionally parameterized convolutions for efficient inference. In *NeurIPS*, 2019.

C. Zhang, S. Bengio, M. Hardt, B. Recht, and O. Vinyals. Understanding deep learning requires rethinking generalization. In *ICLR*, 2017.

A Complete results: ResNet-18

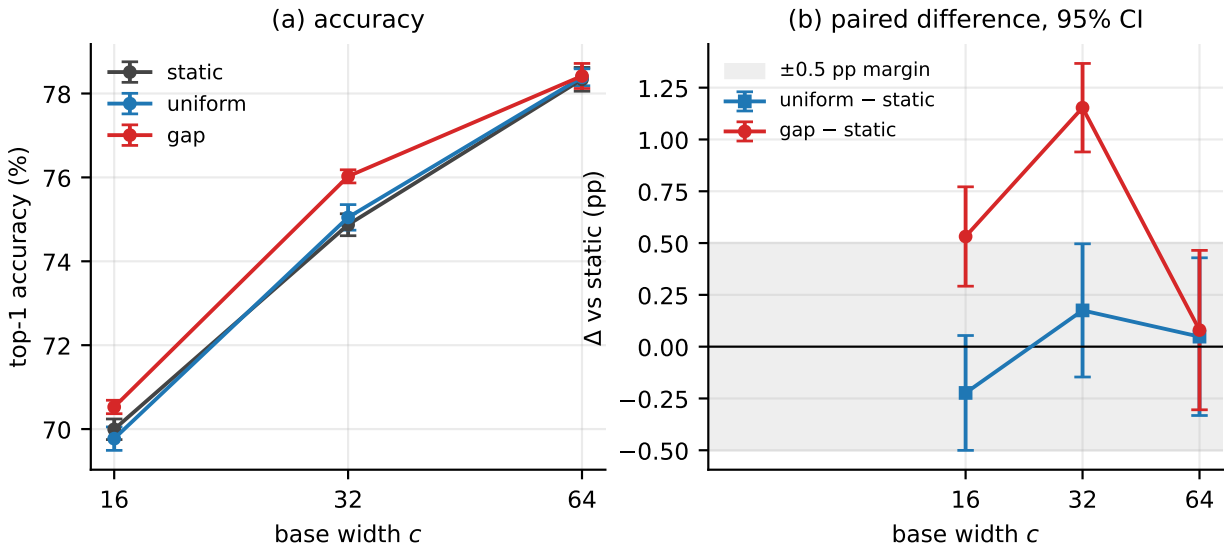


Figure 15: ResNet-18, CIFAR-100. (a) Accuracy against base width, ±1 standard deviation over six seeds. (b) Paired difference from `static` with 95 % confidence intervals.

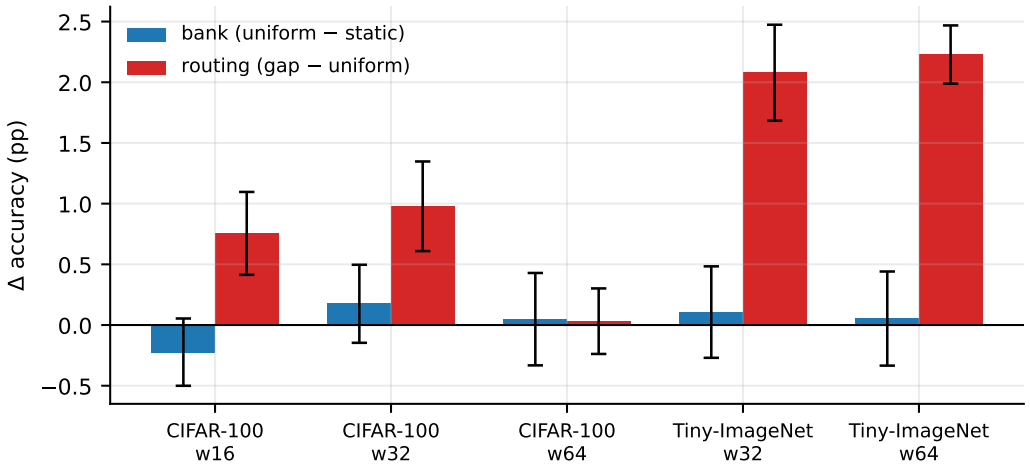


Figure 16: ResNet-18: decomposition of the effect of dynamic convolution into the bank (`uniform − static`) and routing (`gap − uniform`), five conditions, 95 % CIs.

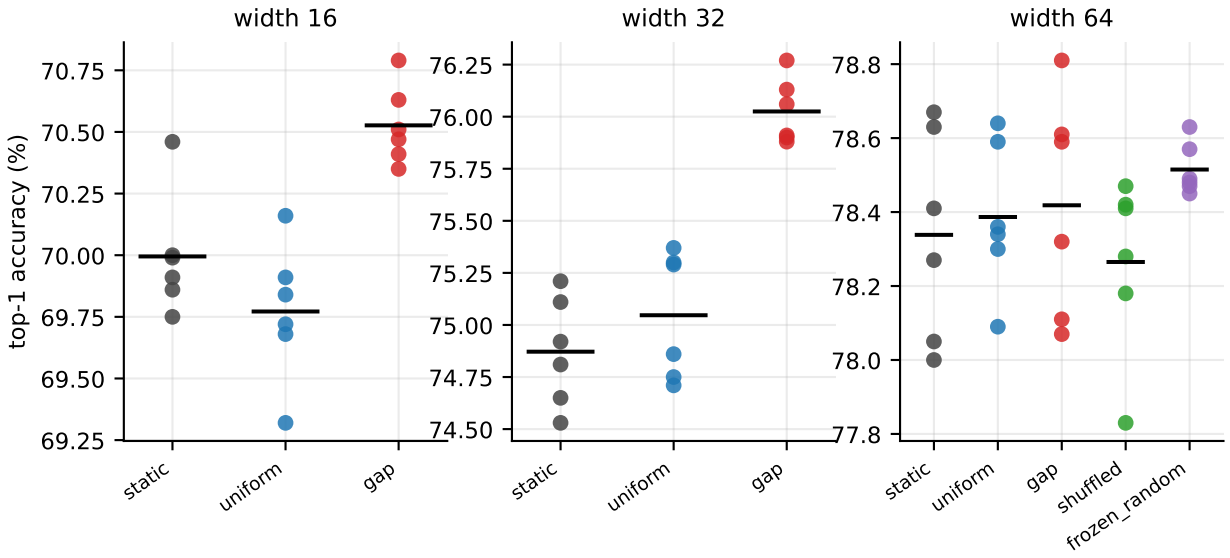


Figure 17: ResNet-18: every CIFAR-100 run at $K = 4$ and $\tau = 1$; bars are arm means.

Table 19: ResNet-18: every run at $K = 4$ and $\tau = 1$: final-epoch top-1 accuracy (%) for each seed, with mean and sample standard deviation.

dataset	width	arm	s0	s1	s2	s3	s4	s5	mean	sd
CIFAR-100	16	`static`	69.91	69.75	69.86	70.46	70.00	69.99	69.995	0.246
CIFAR-100	16	`uniform`	69.68	69.32	69.84	69.91	69.72	70.16	69.772	0.279
CIFAR-100	16	`gap`	70.35	70.51	70.41	70.63	70.79	70.47	70.527	0.160
CIFAR-100	32	`static`	74.81	74.65	74.53	74.92	75.11	75.21	74.872	0.262
CIFAR-100	32	`uniform`	75.29	74.86	74.75	75.30	74.71	75.37	75.047	0.305
CIFAR-100	32	`gap`	76.13	75.88	75.91	75.90	76.27	76.06	76.025	0.156
CIFAR-100	64	`static`	78.05	78.67	78.63	78.27	78.00	78.41	78.338	0.284
CIFAR-100	64	`uniform`	78.64	78.59	78.34	78.30	78.36	78.09	78.387	0.202
CIFAR-100	64	`gap`	78.59	78.81	78.11	78.61	78.07	78.32	78.418	0.299
CIFAR-100	64	`shuffled`	78.42	77.83	78.18	78.41	78.47	78.28	78.265	0.238
CIFAR-100	64	`frozen_random`	78.47	78.49	78.48	78.63	78.45	78.57	78.515	0.070
Tiny-ImageNet	32	`static`	61.93	61.61	61.57	61.46	61.27	61.52	61.560	0.217
Tiny-ImageNet	32	`uniform`	61.52	61.53	61.55	61.62	61.66	62.12	61.667	0.229
Tiny-ImageNet	32	`gap`	63.53	64.20	63.61	63.58	63.91	63.64	63.745	0.260
Tiny-ImageNet	64	`static`	66.29	66.59	65.88	65.97	66.63	66.23	66.265	0.309
Tiny-ImageNet	64	`uniform`	66.27	66.27	66.41	66.21	66.22	66.53	66.318	0.126
Tiny-ImageNet	64	`gap`	68.73	68.35	68.42	68.55	68.71	68.52	68.547	0.152

Table 20: ResNet-18: paired comparisons against `static`, $n = 6$. p_W is the Wilcoxon signed-rank test, which cannot fall below 0.031 at this n.

dataset	width	arm	Δ (pp)	95 % CI	p_t	p_W	d_z	p_{TOST}	verdict
CIFAR-100	16	`uniform`	-0.223	[-0.500, +0.054]	0.0931	0.0938	-0.85	0.0251	equivalent
CIFAR-100	16	`gap`	+0.532	[+0.292, +0.771]	0.0023	0.0312	+2.33	0.6260	higher
CIFAR-100	32	`uniform`	+0.175	[-0.146, +0.496]	0.2206	0.3125	+0.57	0.0242	equivalent
CIFAR-100	32	`gap`	+1.153	[+0.940, +1.367]	$< 10^{-4}$	0.0312	+5.67	0.9997	higher
CIFAR-100	64	`uniform`	+0.048	[-0.332, +0.429]	0.7573	0.8438	+0.13	0.0142	equivalent
CIFAR-100	64	`gap`	+0.080	[-0.305, +0.465]	0.6158	0.5625	+0.22	0.0188	equivalent
CIFAR-100	64	`shuffled`	-0.073	[-0.602, +0.456]	0.7361	1.0000	-0.15	0.0464	equivalent
CIFAR-100	64	`frozen_random`	+0.177	[-0.121, +0.474]	0.1874	0.2188	+0.62	0.0191	equivalent
Tiny-ImageNet	32	`uniform`	+0.107	[-0.270, +0.483]	0.4994	0.6875	+0.30	0.0218	equivalent
Tiny-ImageNet	32	`gap`	+2.185	[+1.781, +2.589]	$< 10^{-4}$	0.0312	+5.67	0.9999	higher
Tiny-ImageNet	64	`uniform`	+0.053	[-0.334, +0.441]	0.7380	1.0000	+0.14	0.0157	equivalent
Tiny-ImageNet	64	`gap`	+2.282	[+1.952, +2.611]	$< 10^{-4}$	0.0312	+7.26	1.0000	higher

Table 21: ResNet-18: configuration and cost. Router parameters are in thousands and are included in the stored total. Training loss is the final-epoch mean over six seeds.

dataset	width	arm	params (M)	router (K)	train loss	min/run
CIFAR-100	16	`static`	0.71	0.0	0.1643	9.6
CIFAR-100	16	`uniform`	2.77	0.0	0.1653	11.4
CIFAR-100	16	`gap`	2.79	21.3	0.0638	33.7
CIFAR-100	32	`static`	2.82	0.0	0.0170	8.6
CIFAR-100	32	`uniform`	11.06	0.0	0.0170	12.9
CIFAR-100	32	`gap`	11.13	74.6	0.0132	39.2
CIFAR-100	64	`static`	11.22	0.0	0.0080	18.8
CIFAR-100	64	`uniform`	44.18	0.0	0.0081	25.2
CIFAR-100	64	`gap`	44.46	288.0	0.0074	86.0
CIFAR-100	64	`shuffled`	44.46	288.0	0.0083	92.3
CIFAR-100	64	`frozen_random`	44.18	0.0	0.0077	85.4
Tiny-ImageNet	32	`static`	2.85	0.0	0.1194	54.2
Tiny-ImageNet	32	`uniform`	11.09	0.0	0.1200	56.3
Tiny-ImageNet	32	`gap`	11.16	74.6	0.0420	284.2
Tiny-ImageNet	64	`static`	11.27	0.0	0.0210	129.0
Tiny-ImageNet	64	`uniform`	44.23	0.0	0.0220	141.0
Tiny-ImageNet	64	`gap`	44.52	288.0	0.0190	687.2

Table 22: ResNet-18: routing entropy by arm, CIFAR-100 width 64, per-seed means.

arm	mean $\tilde{H}$	min	max	$1 - \tilde{H}$
`uniform`	1.0000	1.0000	1.0000	0.00000
`gap`	0.9885	0.9867	0.9912	0.01147
`shuffled`	0.9996	0.9995	0.9997	0.00040
`frozen_random`	0.9971	0.9967	0.9977	0.00287

Table 23: Every ResNet-18 run. GPU-hours are from logged wall-clock time.

dataset	width	K	τ	arms	runs	GPU-h
CIFAR-100	16	4	constant	`static`, `uniform`, `gap`	18	5.5
CIFAR-100	32	4	constant	`static`, `uniform`, `gap`	18	6.1
CIFAR-100	64	4	constant	`static`, `uniform`, `gap`, `shuffled`, `frozen_random`	30	30.8
CIFAR-100	32	2	constant	`gap`	6	2.8
CIFAR-100	32	8	constant	`gap`	6	6.9
CIFAR-100	16	4	annealed	`gap`	6	4.4
CIFAR-100	32	4	annealed	`gap`	6	3.8
CIFAR-100	64	4	annealed	`gap`	6	8.6
Tiny-ImageNet	32	4	constant	`static`, `uniform`, `gap`	18	39.5
Tiny-ImageNet	64	4	constant	`static`, `uniform`, `gap`	18	95.7
total					132	204.1

B Complete results: MobileNetV2

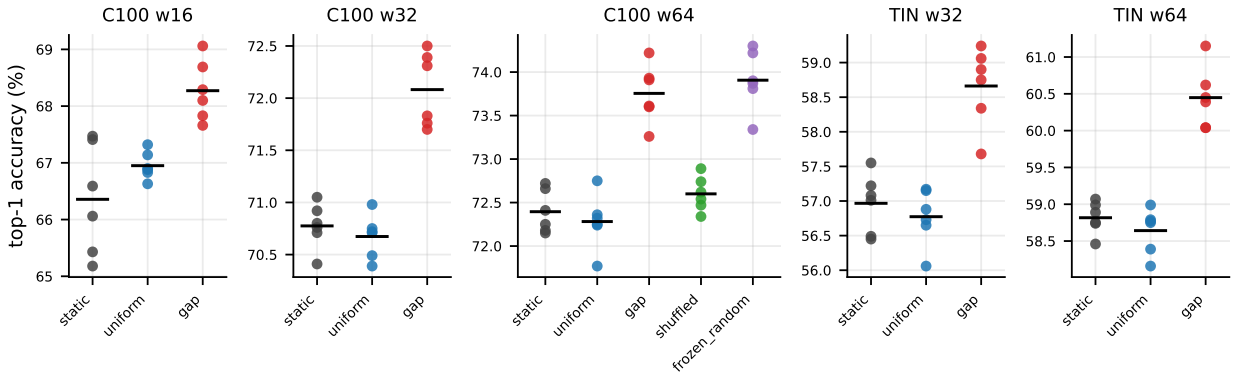


Figure 18: Every MobileNetV2 run at $K = 4$ and $\tau = 1$; bars are arm means.

Table 24: MobileNetV2, every run at $K = 4$ and $\tau = 1$: final-epoch top-1 accuracy (%). † marks the run resumed after a GPU fault.

dataset	width	arm	s0	s1	s2	s3	s4	s5	mean	sd
CIFAR-100	16	`static`	65.43	67.41	66.59	65.18	67.47	66.06	66.357	0.973
CIFAR-100	16	`uniform`	66.88	66.90	66.63	67.14	67.32	66.83	66.950	0.244
CIFAR-100	16	`gap`	68.69	67.83	67.66	69.06	68.10	68.29	68.272	0.528
CIFAR-100	32	`static`	70.92	71.05	70.71	70.80	70.41	70.76	70.775	0.217
CIFAR-100	32	`uniform`	70.49	70.39	70.98	70.72	70.75	70.71	70.673	0.208
CIFAR-100	32	`gap`	72.39	71.76	72.31	71.70	72.50	71.83	72.082	0.356
CIFAR-100	64	`static`	72.66	72.18	72.15	72.25	72.41	72.72	72.395	0.246
CIFAR-100	64	`uniform`	72.32	71.77	72.75	72.24	72.25	72.36	72.282	0.314
CIFAR-100	64	`gap`	74.22	73.26	73.93	73.91	73.60	73.61	73.755	0.335
CIFAR-100	64	`shuffled`	72.34	72.62	72.54	72.74	72.89	72.47	72.600	0.196
CIFAR-100	64	`frozen_random`	74.22	74.30	73.81	73.34	73.87	73.90	73.907	0.342
Tiny-ImageNet	32	`static`	57.01	57.08	56.45	57.22	56.49	57.55	56.967	0.427
Tiny-ImageNet	32	`uniform`	56.06	56.65	56.88	56.73	57.17	57.15	56.773	0.409
Tiny-ImageNet	32	`gap`	58.90	58.34	59.06	57.68	58.75	59.24	58.662	0.570
Tiny-ImageNet	64	`static`	59.07	58.99	58.89	58.74	58.75	58.46	58.817	0.218
Tiny-ImageNet	64	`uniform`	58.39	58.16	58.99	58.75	58.77	58.79	58.642	0.306
Tiny-ImageNet	64	`gap`	60.45	61.15†	60.04	60.62	60.04	60.39	60.448	0.414

Table 25: MobileNetV2: paired comparisons against `static`, $n = 6$.

dataset	width	arm	Δ (pp)	95 % CI	p_t	p_W	d_z	p_{TOST}	verdict
CIFAR-100	16	`uniform`	+0.593	[-0.425, +1.612]	0.1945	0.3125	+0.61	0.5884	inconclusive
CIFAR-100	16	`gap`	+1.915	[+0.404, +3.426]	0.0225	0.0312	+1.33	0.9694	higher
CIFAR-100	32	`uniform`	-0.102	[-0.510, +0.306]	0.5501	0.5625	-0.26	0.0269	equivalent
CIFAR-100	32	`gap`	+1.307	[+0.771, +1.843]	0.0015	0.0312	+2.56	0.9941	higher
CIFAR-100	64	`uniform`	-0.113	[-0.512, +0.285]	0.4978	0.4375	-0.30	0.0275	equivalent
CIFAR-100	64	`gap`	+1.360	[+0.986, +1.734]	0.0002	0.0312	+3.82	0.9990	higher
CIFAR-100	64	`shuffled`	+0.205	[-0.196, +0.606]	0.2456	0.1562	+0.54	0.0585	inconclusive
CIFAR-100	64	`frozen_random`	+1.512	[+1.123, +1.900]	0.0002	0.0312	+4.08	0.9994	higher
Tiny-ImageNet	32	`uniform`	-0.193	[-0.842, +0.455]	0.4782	0.6250	-0.31	0.1393	inconclusive
Tiny-ImageNet	32	`gap`	+1.695	[+0.895, +2.495]	0.0028	0.0312	+2.22	0.9939	higher
Tiny-ImageNet	64	`uniform`	-0.175	[-0.664, +0.314]	0.4001	1.0000	-0.38	0.0742	inconclusive
Tiny-ImageNet	64	`gap`	+1.632	[+1.201, +2.062]	0.0002	0.0312	+3.98	0.9995	higher

Table 26: MobileNetV2: departure of each dynamic layer's router from uniform, $10^3\,(1 - \tilde{H})$, mean over six seeds, $K = 4$. Layers are numbered from the input; a value of 0.00 is below 5×10^{-6}.

	CIFAR-100, width 64				CIFAR-100, width 16		TIN w64
layer	`gap`	`gap` ann.	`frozen_random`	`shuffled`	`gap`	`gap` ann.	`gap`
1	3.66	4.15	4.23	0.11	7.79	16.51	1.87
2	9.38	7.63	6.45	0.11	7.14	0.43	5.87
3	9.54	0.00	2.46	0.01	7.31	0.00	9.59
4	9.36	0.00	3.89	0.02	7.64	2.05	4.22
5	6.34	0.00	6.19	0.02	0.37	0.00	0.90
6	5.33	0.00	2.46	0.00	1.37	0.00	1.65
7	26.08	0.00	4.94	0.02	9.49	0.00	7.38
8	13.15	0.00	2.26	0.02	6.17	0.00	0.04
9	9.26	0.00	1.80	0.00	0.00	0.00	4.38
10	39.87	0.00	2.52	0.00	0.00	0.00	4.37
11	29.20	0.00	3.18	0.27	19.95	0.00	15.63
12	13.11	0.00	5.60	0.04	5.12	0.00	15.31
13	5.94	0.00	1.68	0.00	4.54	0.00	15.55
14	23.36	0.00	0.29	0.03	35.60	0.00	15.25
15	8.10	0.00	1.40	0.20	7.85	0.00	22.07
16	2.89	0.00	0.43	0.07	6.86	0.00	13.50
17	9.83	0.00	0.24	0.00	15.80	0.00	5.21
mean	13.20	0.69	2.94	0.05	8.41	1.12	8.40

Table 27: MobileNetV2: configuration and cost. Parameters are counted from the model definition; router parameters are in thousands (the frozen router is held as buffers and not counted). Training loss is the final-epoch mean over six seeds; minutes are logged wall-clock time.

dataset	width	arm	params (M)	router (K)	train loss	min/run
CIFAR-100	16	`static`	0.37	0.0	0.729	16.2
CIFAR-100	16	`uniform`	0.42	0.0	0.717	20.2
CIFAR-100	16	`gap`	0.49	75.3	0.655	41.0
CIFAR-100	32	`static`	0.82	0.0	0.409	18.0
CIFAR-100	32	`uniform`	0.91	0.0	0.418	21.6
CIFAR-100	32	`gap`	1.20	289.1	0.344	47.3
CIFAR-100	64	`static`	2.35	0.0	0.248	24.8
CIFAR-100	64	`uniform`	2.54	0.0	0.231	26.9
CIFAR-100	64	`gap`	3.68	1137.3	0.168	78.1
CIFAR-100	64	`shuffled`	3.68	1137.3	0.229	92.0
CIFAR-100	64	`frozen_random`	2.54	0.0	0.175	77.4
Tiny-ImageNet	32	`static`	0.94	0.0	1.345	116.1
Tiny-ImageNet	32	`uniform`	1.04	0.0	1.359	114.5
Tiny-ImageNet	32	`gap`	1.33	289.1	1.220	340.6
Tiny-ImageNet	64	`static`	2.48	0.0	1.138	187.4
Tiny-ImageNet	64	`uniform`	2.67	0.0	1.126	191.3
Tiny-ImageNet	64	`gap`	3.81	1137.3	0.995	792.5

Table 28: Every MobileNetV2 run. GPU-hours are from logged wall-clock time.

dataset	width	K	τ	arms	runs	GPU-h
CIFAR-100	16	4	constant	`static`, `uniform`, `gap`	18	7.7
CIFAR-100	32	4	constant	`static`, `uniform`, `gap`	18	8.7
CIFAR-100	64	4	constant	`static`, `uniform`, `gap`, `shuffled`, `frozen_random`	30	29.9
CIFAR-100	32	2	constant	`gap`	6	5.2
CIFAR-100	32	8	constant	`gap`	6	9.9
CIFAR-100	16	4	annealed	`gap`	6	4.1
CIFAR-100	32	4	annealed	`gap`	6	4.7
CIFAR-100	64	4	annealed	`gap`	6	9.1
Tiny-ImageNet	32	4	constant	`static`, `uniform`, `gap`	18	57.1
Tiny-ImageNet	64	4	constant	`static`, `uniform`, `gap`	18	117.1
total					132	253.6